Chapter 6 Wastewater Treatment

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Expected leaning outcomes for Chapter 6

- Wastewater treatment
  - Wastewater sources, constituents, characteristics
  - Wastewater treatment levels & methods
  - Microbial kinetics
  - Biological treatment (suspended growth, attached growth, and hybrid growth)
  - Sludge treatment
  - Nutrient removal

Water treatment systems

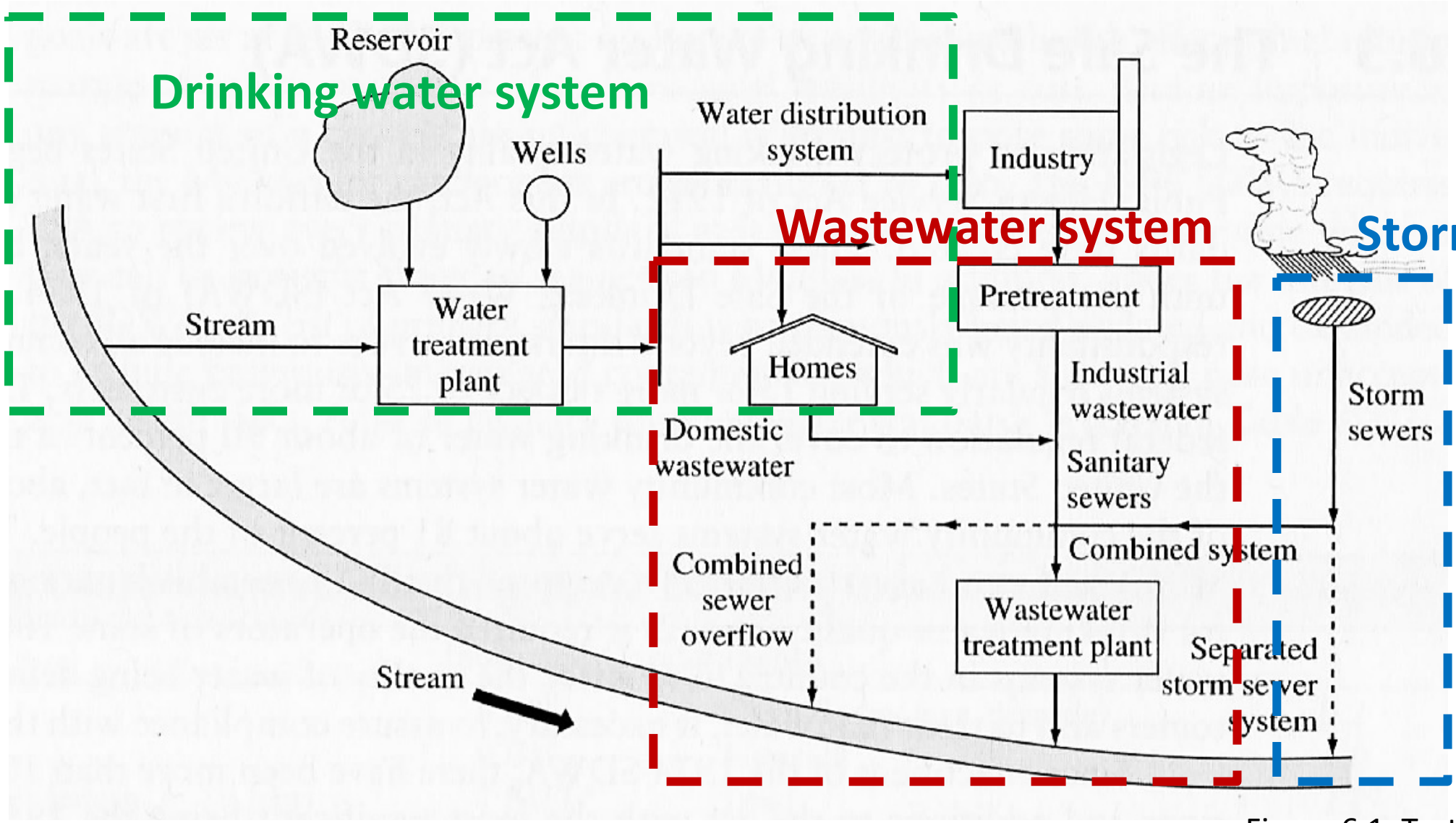


Figure 6.1, Textbook, p283

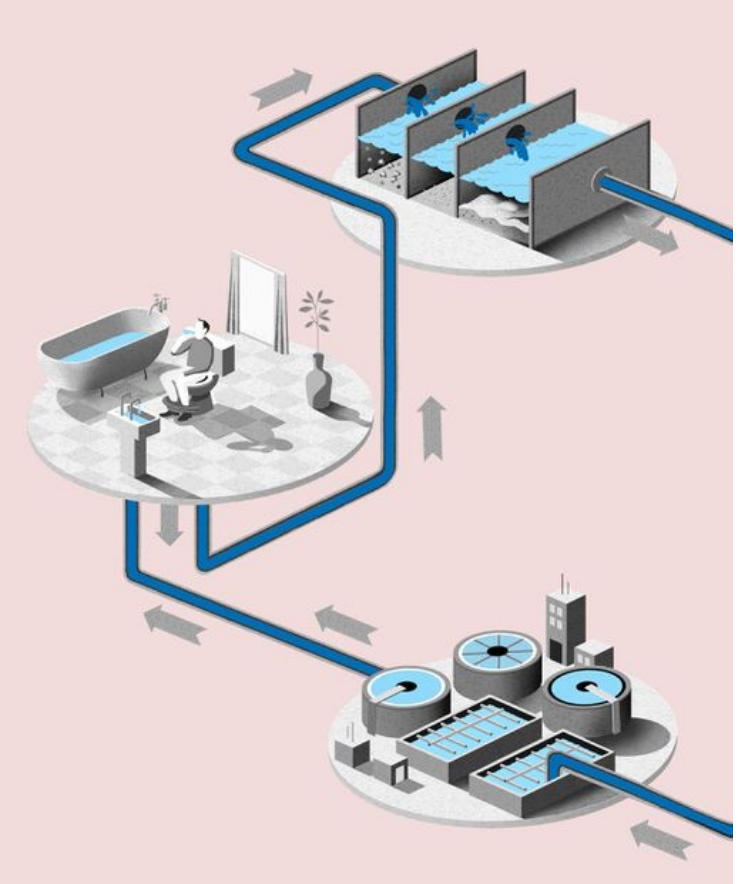
History of wastewater treatment





Direct discharge of wastewater on land or in streams and rivers.

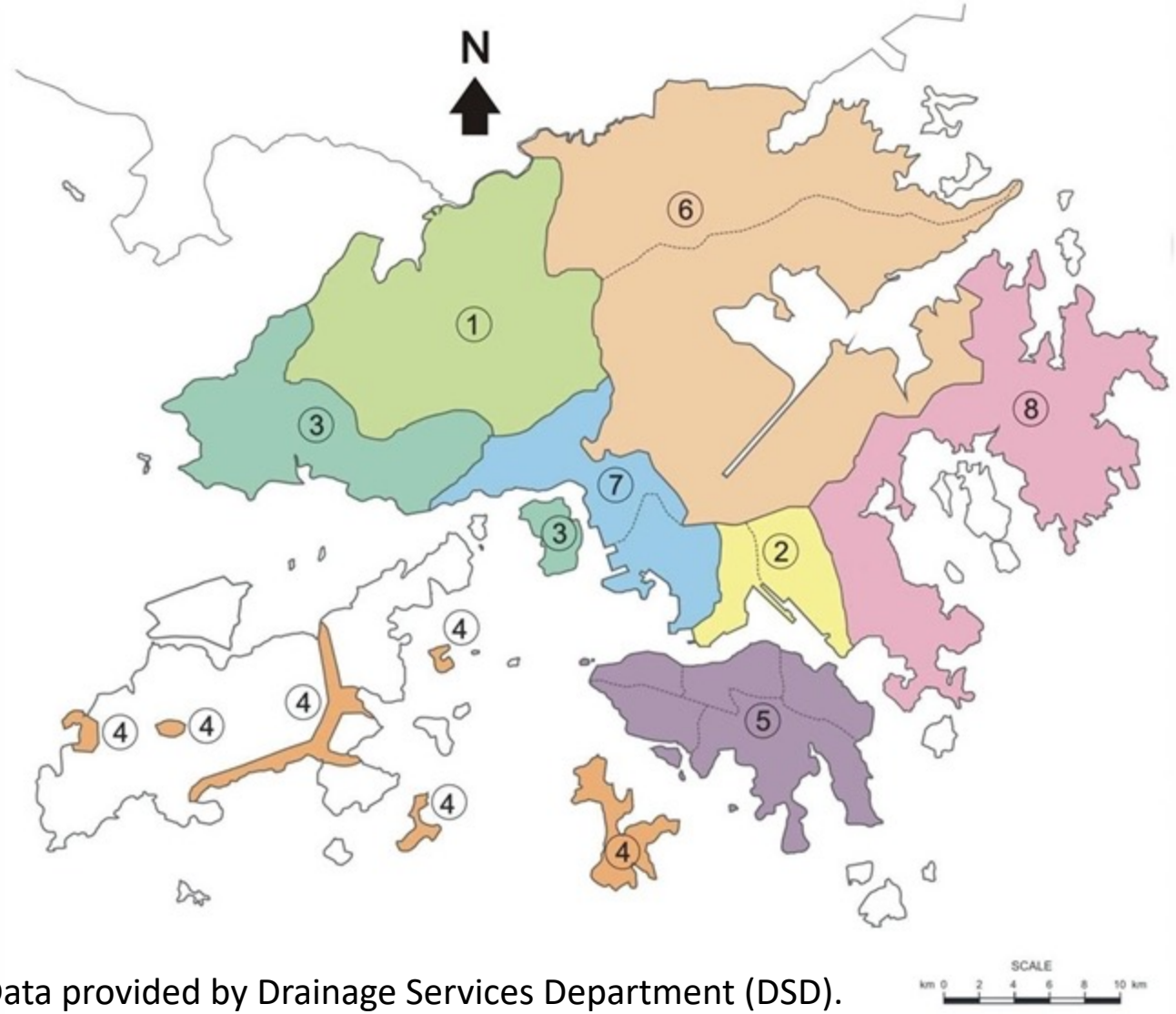
History of wastewater treatment





- Great invention of wastewater collection and treatment system!
- By 1920s, treatment plants were designed and constructed for proper treatment of wastewater prior to disposal.

Wastewater treatment in HK



Data provided by Drainage Services Department (DSD).

- 16 Sewerage Master Plans (SMPs) established at 1989.
- About 93% of the population are now served by the wastewater treatment system.
- 1,700 km wastewater collection network and ~300 wastewater pumping stations and treatment facilities.
- Treating **2.8 million m^3/day** of wastewater.

Source of wastewater

- **Domestic or municipal wastewater**
  - wastewater discharged from residences, institutions such as schools and hospitals, and commercial facilities such as restaurants, shopping malls, etc.
- **Industrial wastewater**
  - wastewater discharged from industrial processes, e.g. pharmaceutical industry, poultry processing.
- **Infiltration and inflow**
  - this includes water that eventually enters the sewer from foundation drains, leaking pipes, submerged manholes, and groundwater infiltration, among others.
- **Stormwater**
  - rainfall runoff and snow melt.

Purpose of wastewater treatment





Overall purpose

- To meet regulatory limits that are protective of public health and the environment.

Specific control targets

(1) the level of solids.

(2) the level of biodegradable organic matter.

(3) the level of pathogens.

(4) the level of toxic compounds in the wastewater.

Wastewater constituents

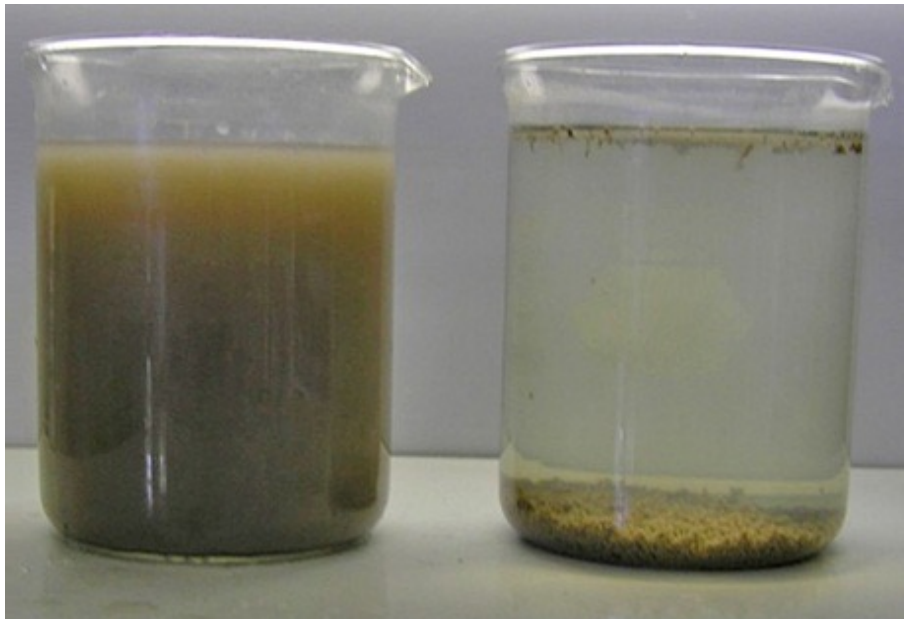
Municipal wastewater

- More than **99.9% water**
- Major constituents are **suspended solids (SS), organic matters and pathogens -> exert oxygen demand (BOD)**

Table 5.1 Environmental impacts of major wastewater pollutants

Pollutant	*Source*	*Environmental impact on receiving waters*
Suspended solids	Municipal wastewater, stormwater	Scum layer on water surface, sludge deposits
Organic matter	Municipal wastewater, possible industrial wastewater	Dissolved oxygen depletion, anaerobic conditions, fish kills
Nutrients	Municipal wastewater, industrial wastewater	Eutrophication and impairment of water quality
Pathogens	Municipal wastewater	Transmission of diseases
Heavy metals	Industrial wastewater	Toxic to aquatic life
Refractory organics	Industrial wastewater	May be toxic or carcinogenic
Endocrine disrupting compounds	Municipal wastewater	Feminization of fish, possible broader scope of impacts

Source: Adapted from Peavy et al. (1985).



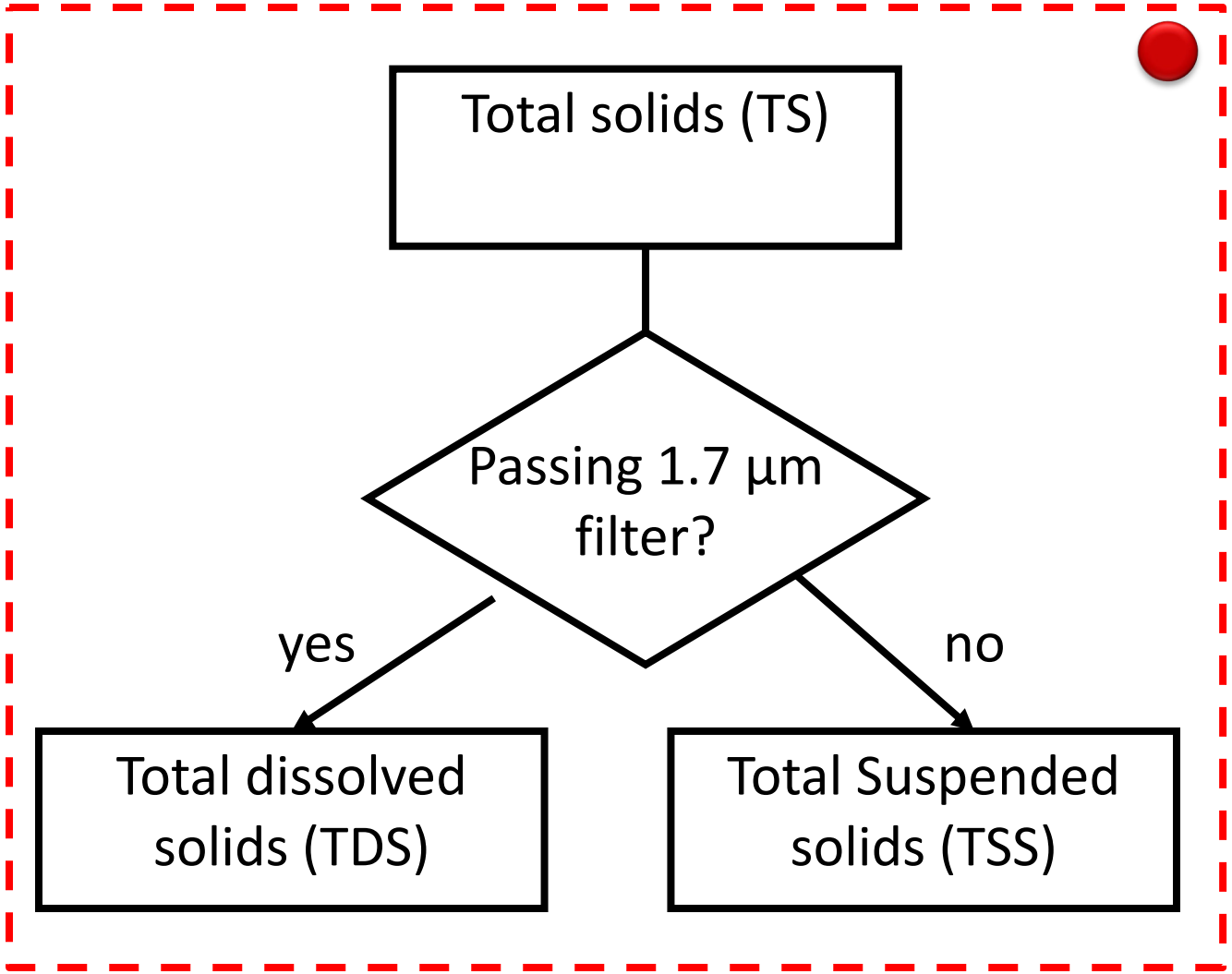
Municipal wastewater characteristics

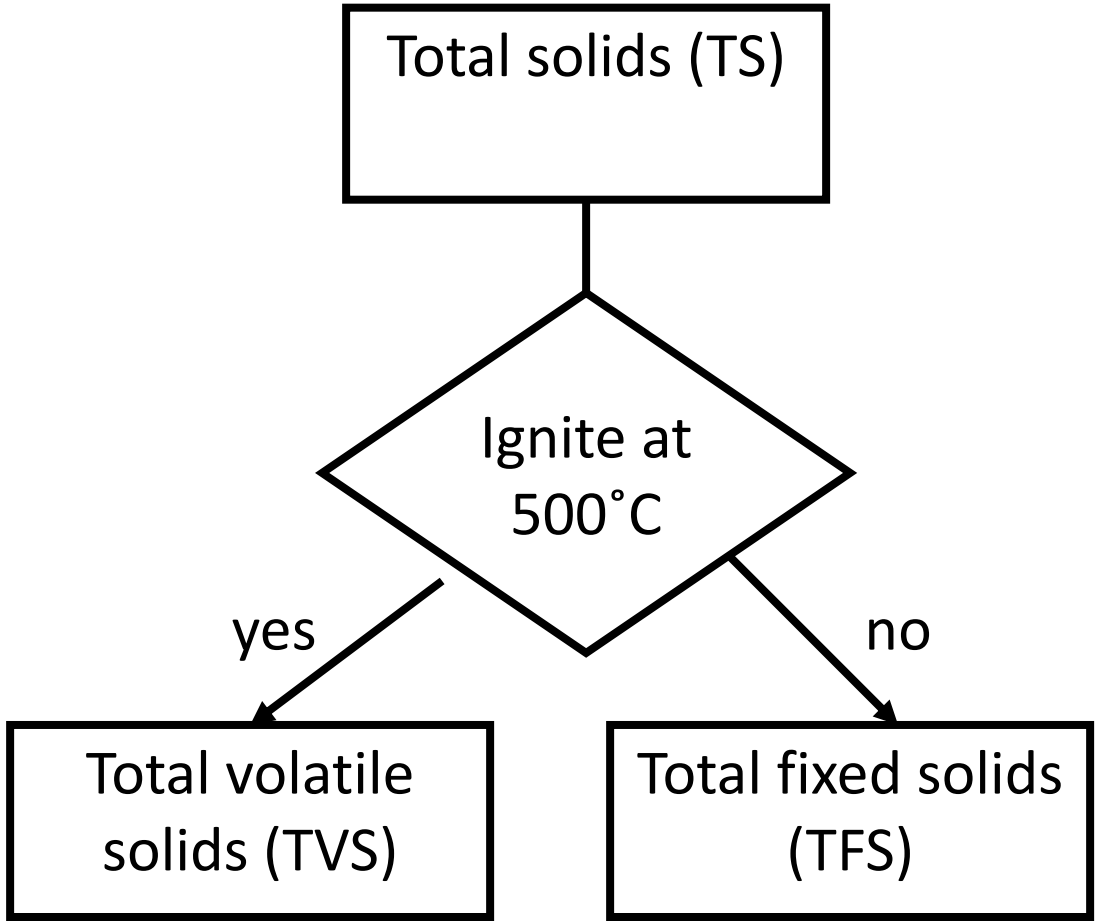
Table 5.2 Typical characteristics of untreated municipal wastewater

Component	*Concentration range*
Biochemical oxygen demand, BOD_5 at 20°C	100–360 mg/L
Chemical oxygen demand, COD	250–1000 mg/L
Total organic carbon, TOC	80–300 mg/L
Total Kjeldahl nitrogen (TKN)	20–85 mg/L as N
Total phosphorus (TP)	5–15 mg/L as P
Oil and grease	50–120 mg/L
Total solids (TS)	400–1200 mg/L
Total dissolved solids (TDS) solids **passing** through a 1.7 µm filter	250–850 mg/L
Total suspended solids (TSS) solids **retained** by a 1.7 µm filter	110–400 mg/L
Volatile suspended solids (VSS)	90–320 mg/L
Fixed suspended solids (FSS)	20–80 mg/L
Settleable solids	5–20 ml/L
Total coliform	10^6–10^{10} No./100 ml
Fecal coliform	10^3–10^8 No./100 ml

Source: Adapted from Metcalf and Eddy (2003).

Municipal wastewater characteristics





Industrial wastewater characteristics

Table 5.3 Typical characteristics of selected industrial wastewaters

Industry	*BOD, mg/L*	*COD, mg/L*	*TSS, mg/L*
Milk processing	1300	3100	300
Meat processing	1400	2500	950
Pulp and paper (kraft)	300	550	250
Tannery	4000	7500	15,000
Slaughterhouse (cattle)	2000	3600	800
Cheese production	3000	5500	950
Pharmaceuticals	280	390	160

Source: Adapted from Hammer and Hammer (2012) and Davis (2011).

Wastewater treatment methods

- **Physical treatment**
  - **Removal of pollutants** from the wastewater by **simple physical forces**.
  - E.g. sedimentation, screening, filtration.
  - Used mainly for **removal of suspended solids**.
- **Chemical treatment**
  - The **addition of chemicals** to achieve **conversion or destruction of contaminants** through chemical reactions.
  - E.g. coagulation–flocculation for solids removal, disinfection for pathogen destruction, chemical precipitation for phosphorus removal.
- **Biological treatment**
  - The **conversion or destruction** of contaminants with the help of **microorganisms**.
  - To remove and reduce the biodegradable organic matter to an acceptable level according to regulatory limits. Biological treatment is also used to remove nutrients such as nitrogen and phosphorus from wastewater.
  - E.g., activated sludge process, membrane bioreactor, trickling filter.

Levels of wastewater treatment

- **Primary treatment**
  - Physical removal of **a portion of the suspended solids** from wastewater, usually by sedimentation
  - Physical processes (screening, skimming, sedimentation)
  - Remove pollutants by size, gravity, or flotation
  - Remove **~ 35% BOD and ~ 60% SS**

- **Enhanced primary treatment**
  - Use of **chemical treatment** to **obtain additional solids removal** in a sedimentation process
  - **Chemical coagulants** are used to promote coagulation and flocculation of solids in a sedimentation tank, resulting in **enhanced suspended solids removal**

Levels of wastewater treatment

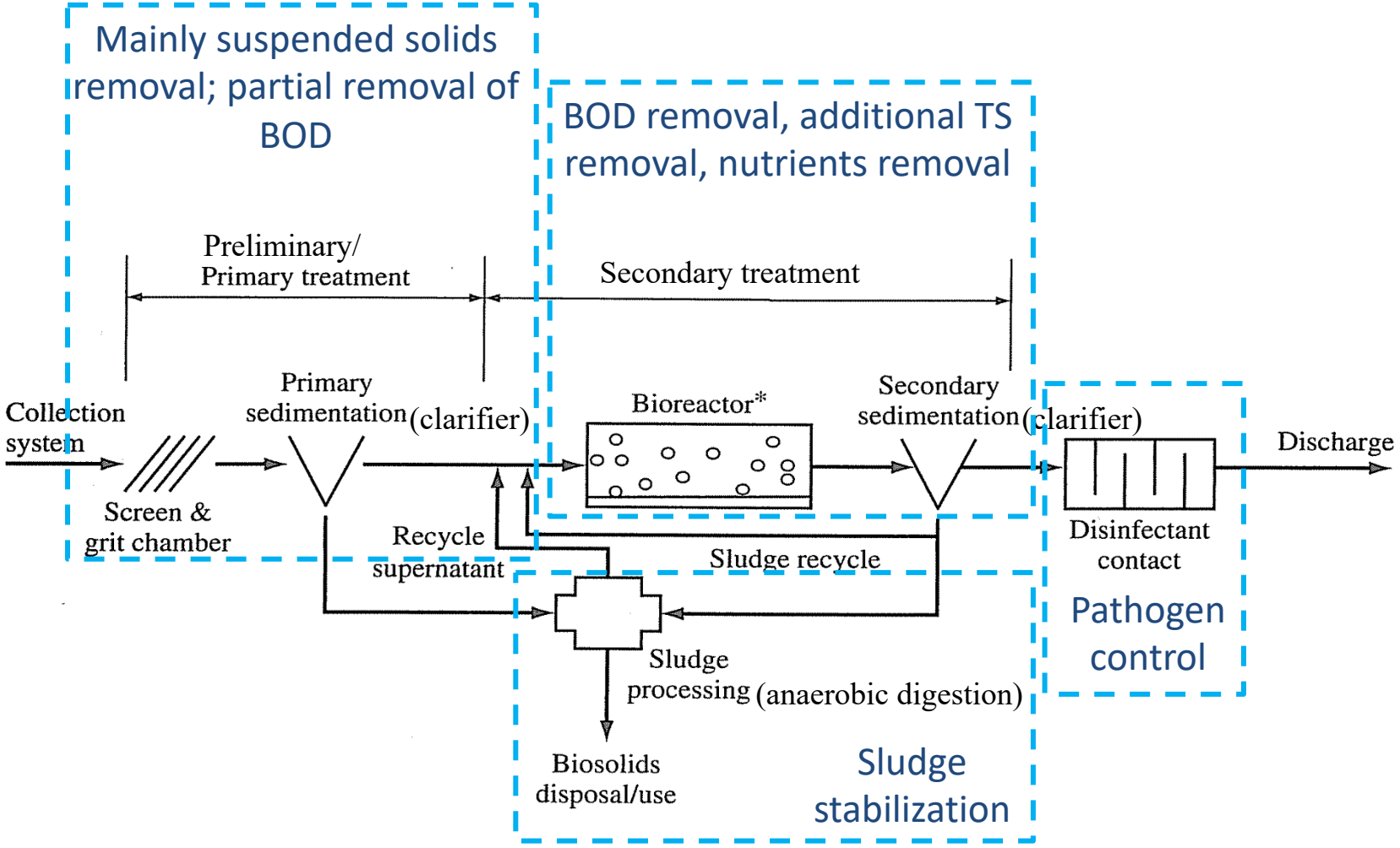
- **Secondary Treatment**
  - Involves **biological treatment** for **degradation of organic matter and solids reduction**
  - Remove **~ 90% BOD and ~ 90% SS**
  - Carried out in a biological reactor followed by a secondary sedimentation (clarifier)
  - E.g., activated sludge process, trickling filter

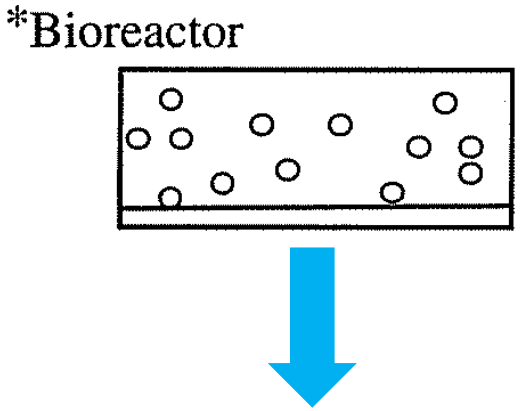
- **Secondary Treatment with Nutrient Removal**
  - **Chemical/biological treatment** for **removal of nutrients,** such as nitrogen and/or phosphorus
  - May require additional reactors or combined with secondary treatment reactor
  - E.g., nitrification-denitrification

Levels of wastewater treatment

- **Tertiary treatment**
  - Primary and secondary treatment processes are not effective in removing nutrients (N & P), dissolved material (metals, salts), and non-biodegradable (refractory substances)
  - A final treatment stage to **further improve the effluent quality** before it is discharged to the receiving environment
  - E.g. granular media filtration used for removal of residual suspended solids, and disinfection for pathogen reduction
  - Additional treatment for nutrient removal
- **Advanced treatment**
  - Used when **additional removal of wastewater constituents** is desired due to **toxicity** of certain compounds, or for potential **water reuse** applications
  - May require additional reactors or combined with secondary treatment reactor
  - E.g. activated carbon adsorption for removal of volatile organic compounds, ion exchange for removal of specific ions, etc.

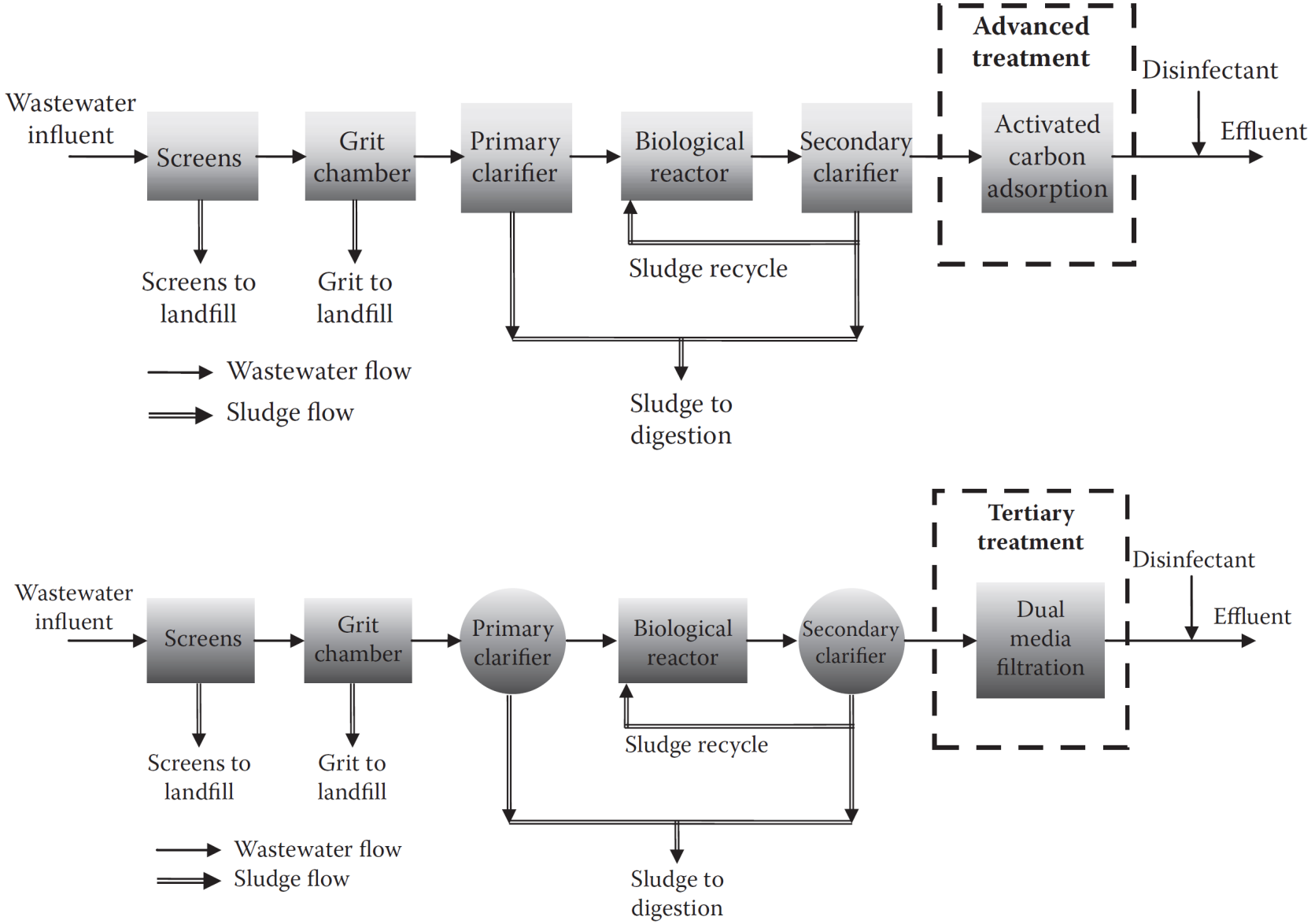
Wastewater treatment scheme



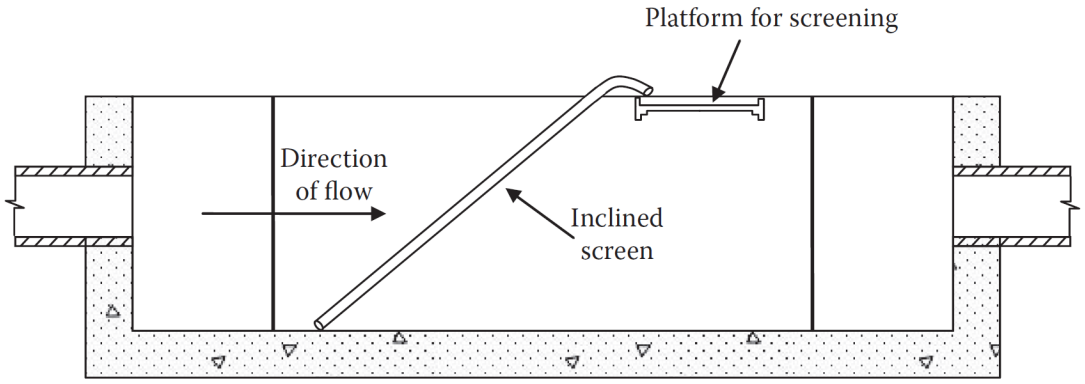


Oxidation ditch
Aeration basin
Membrane bioreactor
Activated sludge
Trickling filter
etc.

Wastewater treatment scheme



Primary treatment: Screen



(a) Section



Trash rack (50 – 150 mm size opening)



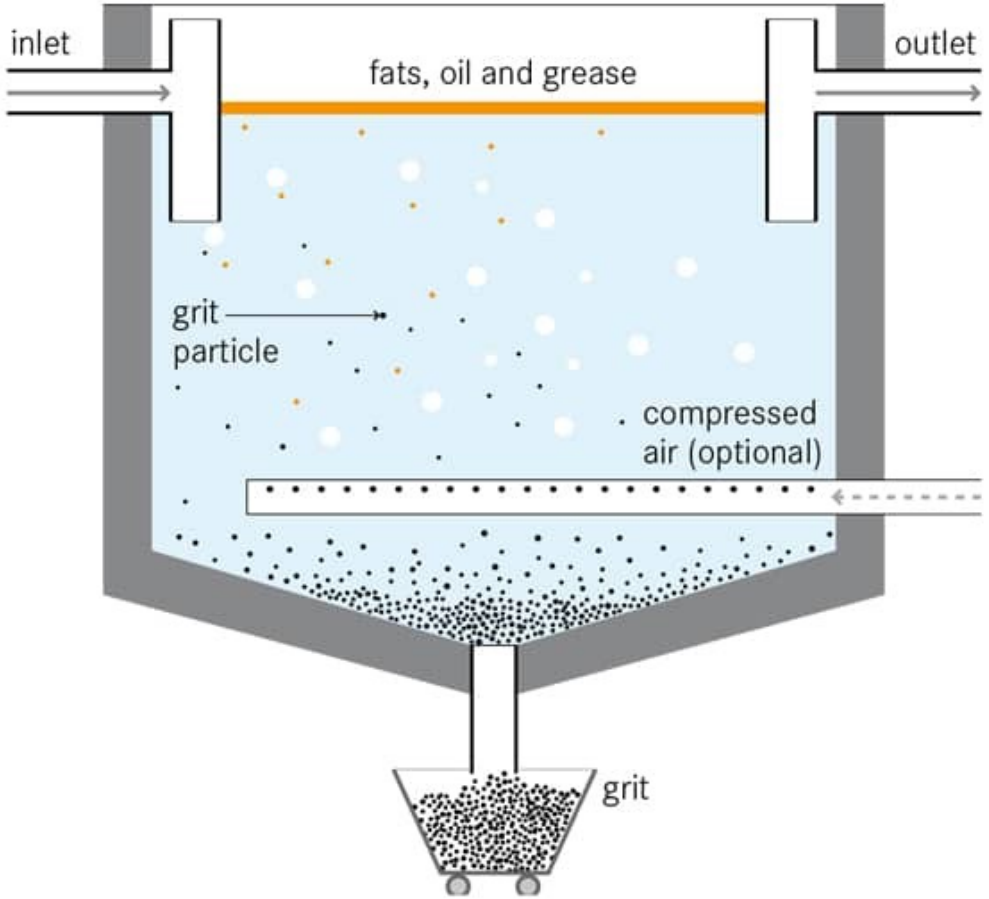
Bar screen (25 – 75 mm size opening)



Fine screen (< 6 mm size opening)

Primary treatment: Grit chamber

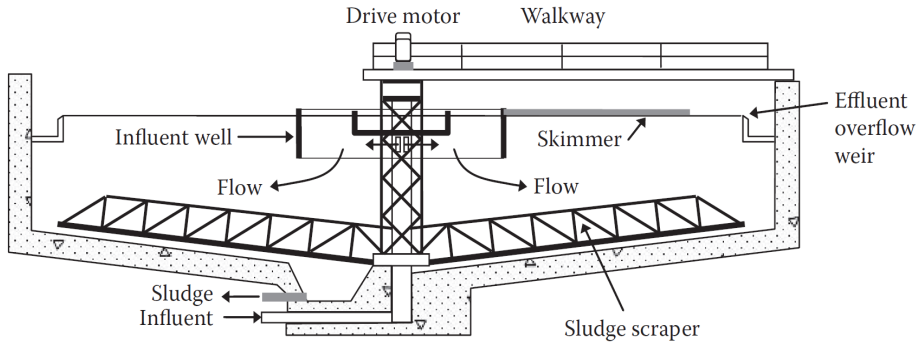
- Grit are sand, gravel, or other mineral material that has a nominal diameter of 0.15–0.20 mm or larger
- Also include ash, wood chips, coffee grounds, egg shells, and other non-putrescible organic matter
- Purpose:
  - Remove materials that may form heavy deposits in pipelines
  - protect pumps and other mechanical equipment from abrasion
  - reduce the frequency of digester cleaning caused by grit accumulation
  - separate heavier inert solids from lighter biodegradable organic solids



Primary treatment: Primary Sedimentation

- Physical removal of a portion of the suspended solids (SS and BOD) from wastewater
- **Soluble pollutants not removed**
- Removal Efficiency of Primary Treatment
  - TSS – 50 to 65%
  - BOD – 25 to 40%







Primary treatment: Primary Sedimentation

- Design of primary sedimentation tanks (primary clarifiers)
  - Surface Overflow Rate (SOR)
    - Defined as the **flow rate (Q) per unit surface area (A)**
      - » V_o = SOR, $m^3/m^2.d$
      - » Q = average daily flow, m^3/d
      - » A = total surface area of tank, m^2

$$V_0 = \frac{Q}{A}$$

- Typical design parameters
  - Depth: 3 – 5 m
  - Surface overflow rate:
    - 30-100 $m^3/m^2.d$ for conventional clarifiers
    - 1000-3000 $m^3/m^2.d$ for high-rate clarifiers
  - HRT: 1.5 – 3 hr

$$\theta = \frac{V}{Q}$$

Primary treatment: Primary Sedimentation

Example 6.8 Sizing a primary clarifier

A town of 30,000 send 0.5 m^3 of wastewater per person per day to the wastewater treatment plant. A conventional circular primary clarifier would be designed to have an average detention time of 2.0 hours and an average overflow rate of 40 m/d, whereas a high-rate clarifier would have an average overflow rate of 1,500 m/d.

a. What would be the dimension of the conventional clarifier?
b. How much area would be saved by using a high-rate clarifier?

Solution

a. At 0.5 m3/person/d, the daily flow rate for 30,000 people would be 15,000 m3/d, so the clarifier surface area is

$$A = \frac{Q}{v_0} = \frac{15{,}000}{40} = 375\ \mathrm{m}^3$$

The tank diameter is $d = \sqrt{\frac{4A}{\pi}} = \sqrt{\frac{4\times375}{3.14}} = \mathbf{22\ m}$

The tank depth is $h = \frac{Q\theta}{A} = \frac{15{,}000\times2}{375\times24} = \mathbf{3.3\ m}$

Primary treatment: Primary Sedimentation

Solution

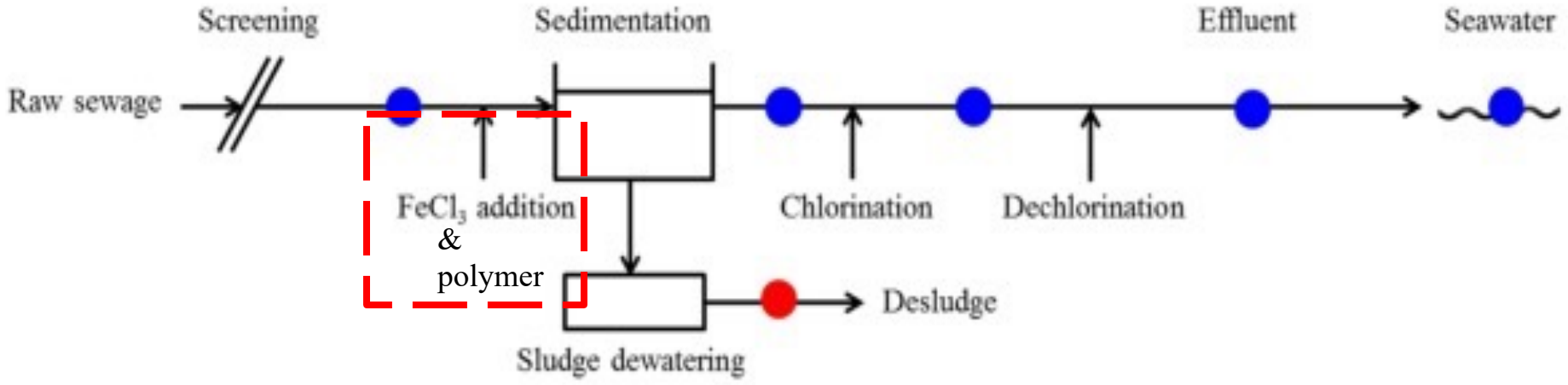
b. For the high-rate clarifier:

$$A = \frac{Q}{v_0} = \frac{15{,}000}{1{,}500} = \mathbf{10\ m^2}$$

Thus, the high-rate clarifier would use about 3% of the land area as the conventional clarifier. In reality, the area saving would not be as large as this, because some additional area would be needed for the coagulant/ballast dosing equipment and for a possible flocculation tank.

Chemically-enhanced primary treatment (CEPT)

- A process that **uses chemicals for coagulation, flocculation, and precipitation of particulate/dissolved solids** in the wastewater **as a primary step in clarification**
- Coagulant: e.g., ferric chloride
- Flocculant: e.g., polymers
- For **phosphate removal**, **clarification of wastewater**, **reduction in sludge volume**, and **increase in surface overflow rate (SOR)**

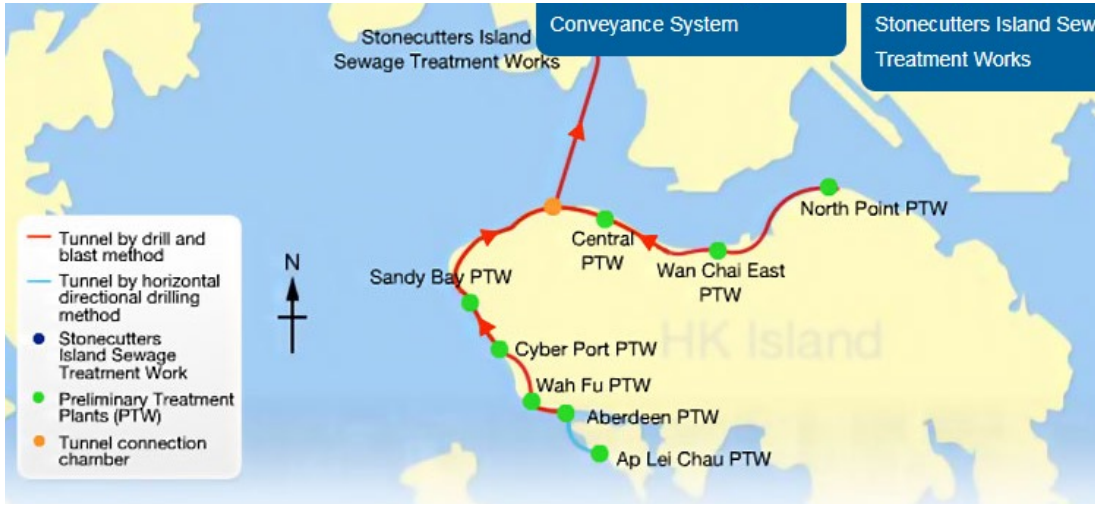


Stonecutters Island wastewater treatment plant using CEPT

Chemically-enhanced primary treatment (CEPT)



Stage 1 - 1.7 million m^3/d



Stage 2A – 2.45 million m^3/d



Stonecutters Island wastewater treatment plant

Data provided by Drainage Services Department (DSD).

Secondary treatment (biological)

- Purposes
  - Remove **BOD**
  - Additional removal of TSS
  - Nearly all pathogens
  - **Nutrients (nitrogen, phosphorus) removal** may also be an objective
  - Carried out in a biological reactor followed by a secondary sedimentation (clarifier)
- Design biological treatment processes :
  - suitable environmental conditions.
  - enhance reaction rates (short treatment time -> smaller reactors -> lower capital/ operating cost, land-saving).
- Requires a thorough understanding of:
  - **microbial growth kinetics**, substrate utilization, principles of **mass balance**, **reactor kinetics**, operational parameters.

Nutrient removal

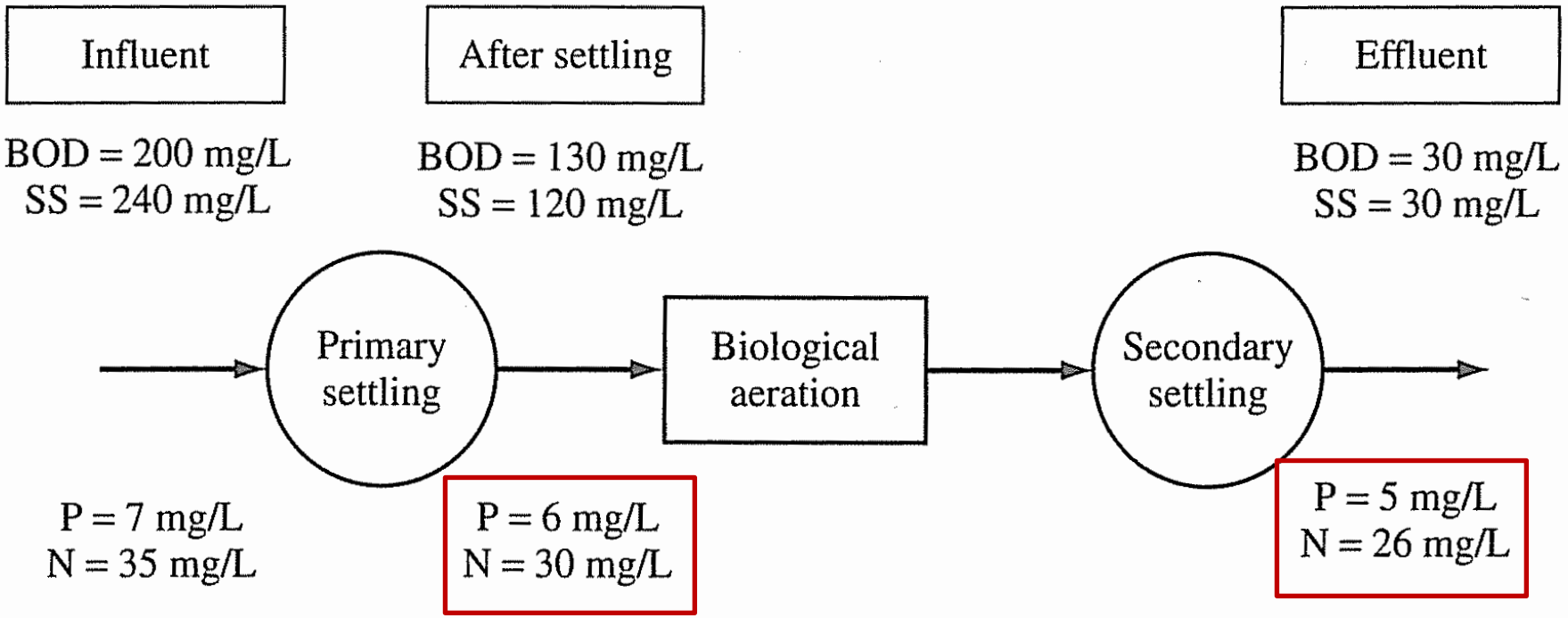


FIGURE 6.16 Approximate concentrations of BOD_5, suspended solids, total nitrogen, and total phosphorus as wastewater passes through a secondary wastewater treatment plant. (*Source:* Based on Hammer and Hammer, 1996.)

Nutrient removal

- Nitrogen removal
  - Prevent eutrophication
  - NH_3 is toxic to fish
  - Removal methods
    - Ammonia stripping
    - Nitrification/denitrification
      - Nitrification: oxidation of ammonia to NO_3^-, sludge age > 15 days
      - Denitrification: convert NO_3^- to nitrogen gas under anoxic condition



Nitrification

Nitrifying bacteria: *Nitrosomonas* / *Nitrobacter*

$$NH_4^+ \xrightarrow[\text{Aerobic } (O_2)]{} NO_2^- \xrightarrow[\text{Aerobic } (O_2)]{} NO_3^-$$

Long retention time of ≥ 15 days needed

Denitrification

$$NO_3^- + \text{organic matter (BOD/COD)} \xrightarrow[\text{Anaerobic (no } O_2)]{\text{Denitrifying bacteria}} N_2 + CO_2 + H_2O$$

Nutrient removal

- Phosphorus removal
  - Prevent eutrophication
  - Removal method: Chemical precipitation
    - Ferric chloride $FeCl_3$
    - Alum $Al_2(SO_4)_3$ → $Al_2(SO_4)_3 + 2PO_4^{3-} \rightarrow 2\ AlPO_4\ (s) + 3\ SO_4^{2-}$
    - Lime $Ca(OH)_2$

Biological Nitrogen Removal Process

Achieved with a two-stage process:

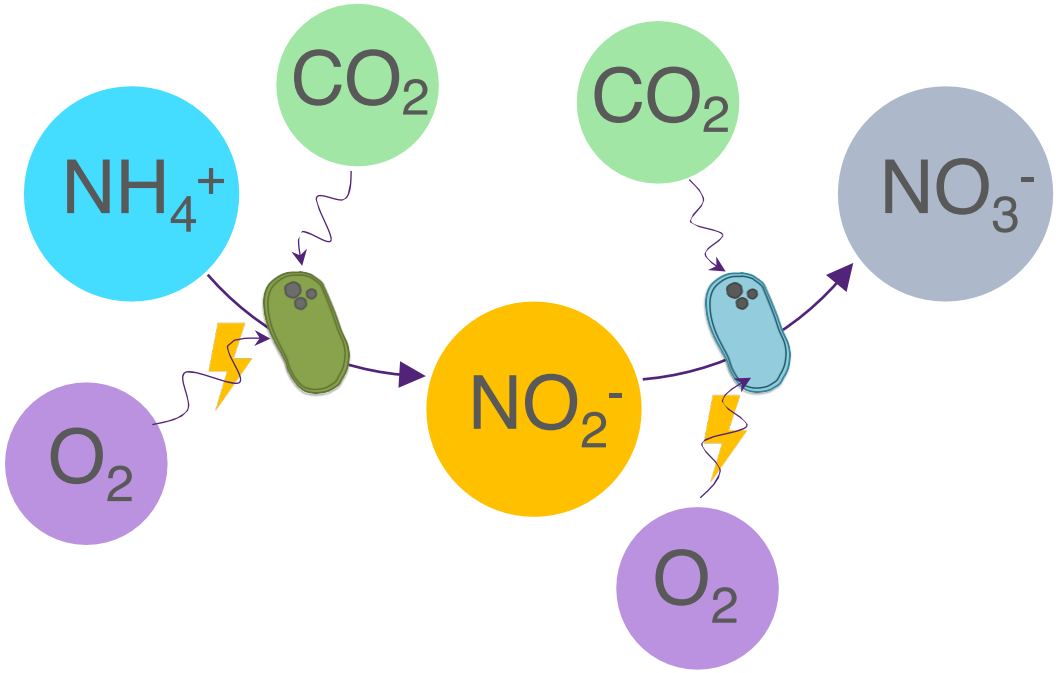
1. **Nitrification** (oxidation of NH_4^+ to NO_3^-)
2. **Denitrification** (reduction of NO_3^- to N_2)

The two processes are performed by two different types of microorganisms:

Organisms	Type of Reaction	Carbon source	Energy source	Electron acceptor	Products
Aerobic Heterotrophs	Aerobic oxidation	Organics	Organics	O_2	CO_2, H_2O
Aerobic Autotrophs	Nitrification	CO_2	NH_4^+, NO_2^-	O_2	NO_2^-, NO_3^-
Facultative Heterotrophs	Anoxic denitrification	Organics	Organics	NO_2^-, NO_3^-	N_2, CO_2, H_2O

Nitrification

- Nitrification is the conversion of ammonium (NH_4^+) into nitrate (NO_3^-) in the presence of O_2. The process is carried out by **autotrophic** bacteria (they use inorganic C) that obtain energy from the oxidation of NH_4^+:



Ammonia-oxidizing bacteria (AOB) (*e.g.*, Nitrosomonas)



Nitrite-oxidizing bacteria (NOB) (*e.g.*, Nitrobacter)



Nitrification

Carried out in 2 steps by **autotrophic** bacteria:

1. $NH_4^+ + 1.5O_2 \rightarrow NO_2^- + 2H^+ + H_2O$ *(Nitrosomonas)*

2. $NO_2^- + 0.5O_2 \rightarrow NO_3^-$ *(Nitrobacter)*



(1+2) $NH_4^+ + 2O_2 \rightarrow NO_3^- + 2H^+ + H_2O$

- *Nitrobacter* grow faster than *Nitrosomonas*, hence, step 1 is normally fast and nitrite does not accumulate (hence the conversion of NH_4^+ is the limiting step);
- Nitrification reactions proceed slowly, require adequate oxygen availability (>0.5 mg/L) and alkalinity (see next slide);
- Nitrifying organisms are sensitive to temperature, pH (preferring near 7), and the presence of toxic chemicals.

Nitrification

- Including biomass formation, the overall nitrification reactions is written as:

  $NH_4^+ + \mathbf{1.863O_2} + 0.098HCO_3^- \rightarrow 0.0196C_5H_7NO_2 + 0.98\ NO_3^- + 0.0941H_2O + 1.98H^+$

- **Oxygen requirements:**

  Including biomass growth, 4.3 g O_2/g NH_4^+-N nitrified are required

  For design, this is normally increased to 4.6 and has to be added to the O_2 demand for the oxidation of organic matter.

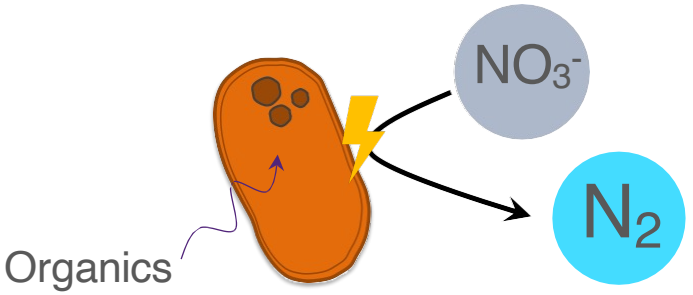
- **Alkalinity used:**

  7.14 g $CaCO_3$/g NH_4^+-N nitrified

- **Growth rate:** μ_{max} = **0.3 – 0.8 d^{-1}** (slower than heterotrophs) (**SRT:** 10-20 d)

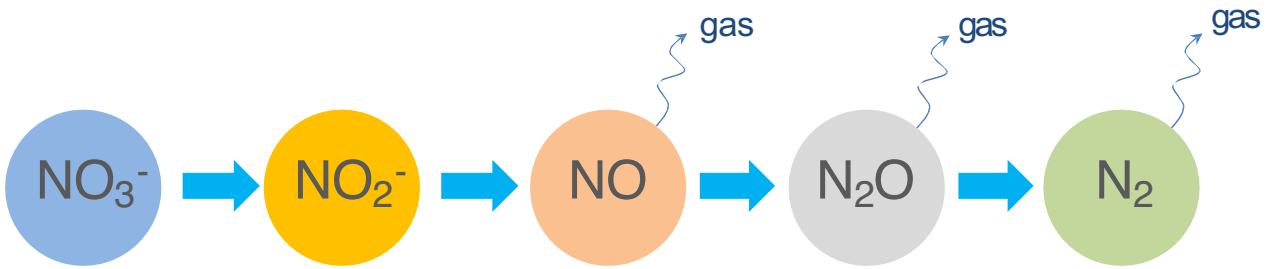
Denitrification

- Conversion of Nitrate to Dinitrogen carried out by **heterotrophic** bacteria (use organic Carbon sources)



- Normally performed by **facultative anaerobic bacteria**, (*e.g.*, of the genera *Paracoccus* and *Pseudomonas*) which in the absence of O_2, utilise **nitrate** as terminal electron acceptor during oxidation of organic matter.

- Denitrification encompasses multiple reduction steps. Incomplete denitrification produces **Nitrous oxide (N_2O)**, which is a potent **greenhouse gas**.



Denitrification and COD requirements

- Assuming biodegradable organic matter in WW can be written as $C_{10}H_{19}O_3N$, the removal of nitrate to dinitrogen can be written as follows:

$$C_{10}H_{19}O_3N + 10NO_3^- \rightarrow 5N_{2(gas)} + 10CO_2 + 3H_2O + NH_3 + 10OH^-$$

- **Carbon demand**:
  2.9 g COD/g NO_3-N denitrified
  ~ 3.5 - 7 g COD/g NO_3-N with biomass
- In principle, COD consumption for denitrification is an advantage as it results in lower requirements for aeration, since part of the organics are consumed for NO_3 reduction.
- Influent **COD/TKN = 11-15** is required to assure complete denitrification

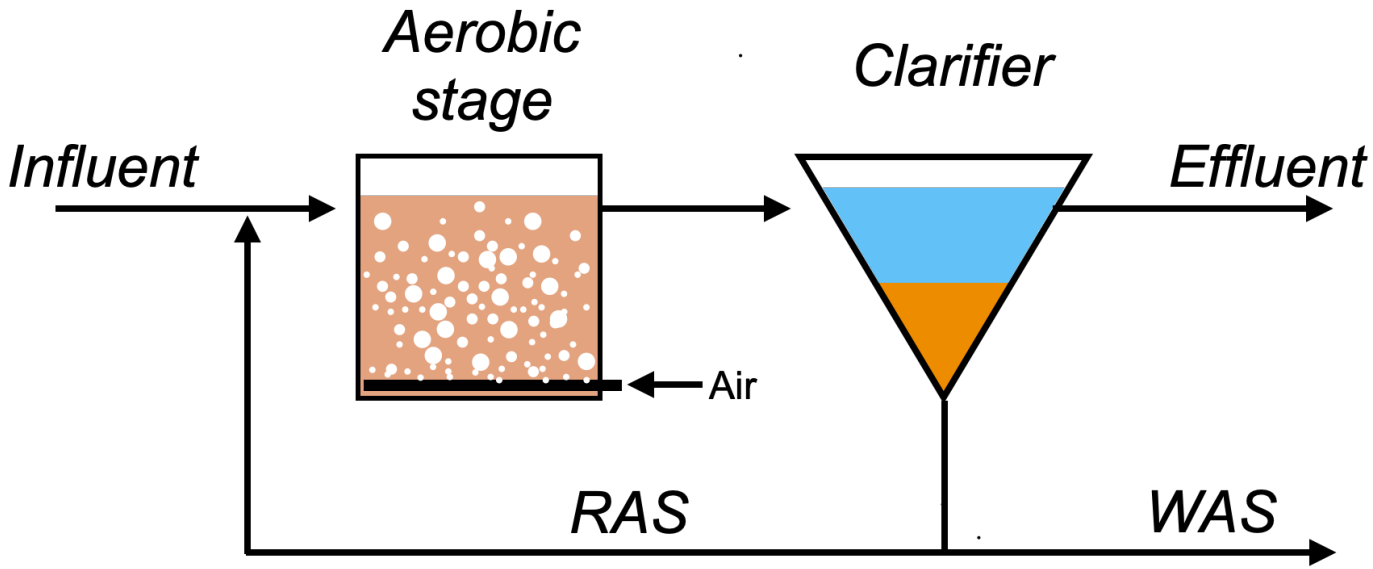
Denitrification and COD requirements

- When the COD available in the wastewater is not sufficient, additional supply is necessary, e.g, *via* Methanol addition:

$$6\ NO_3^- + 5\ CH_3OH \rightarrow 3\ N_2 + 5\ CO_2 + 7\ H_2O + 6\ OH^-$$

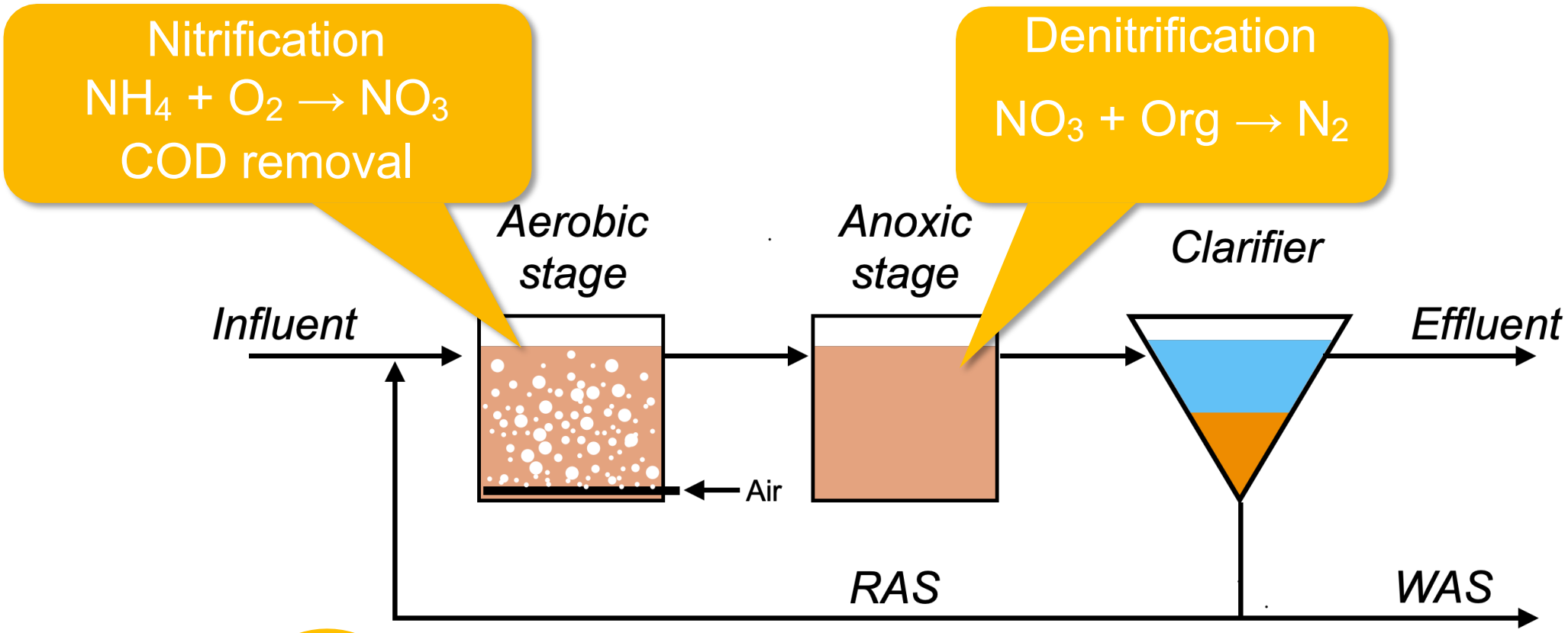
- Denitrification rates: ~2/3 higher with methanol than with available COD (because methanol is readily biodegradable)
- Alkalinity produced:
  3.6 mg $CaCO_3$ /mg NO_3^- - N denitrified
- Growth rate: $\mu_{max} = 2 - 6\ d^{-1}$ (**faster than nitrifiers**)
- Yield coefficient: $Y = 0.18 - 0.4\ g_{VSS}/g_{COD}$

Conventional Activated Sludge process



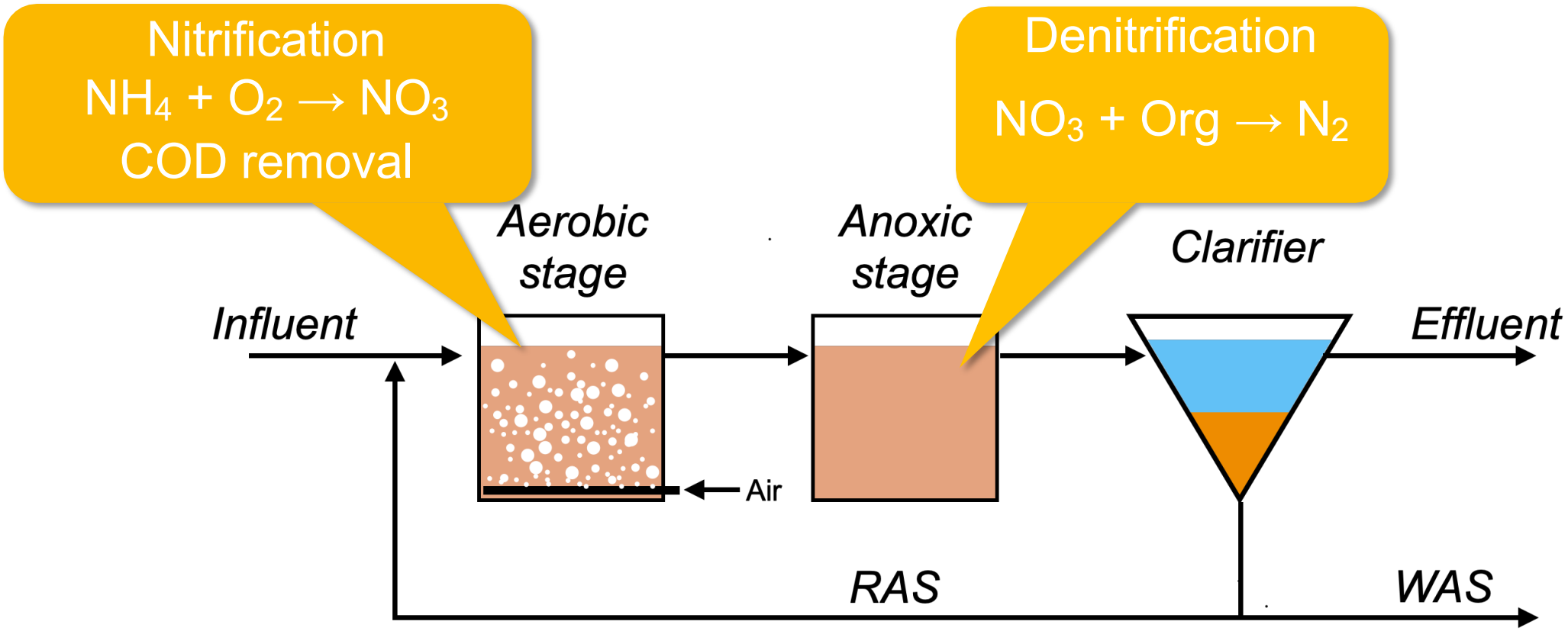
- In the **aerobic stage**, organic matter (BOD) is degraded in the presence of O_2, hence air is added to the system.
- In the **clarifier**, phase separation is used to produce a clarified effluent and a sludge mainly comprising of microorganisms. A portion of it (RAS) is sent to the head of the aerobic stage, while the excess sludge (WAS) is removed and processed for disposal.

Wuhrmann Process Nitrification + Denitrification



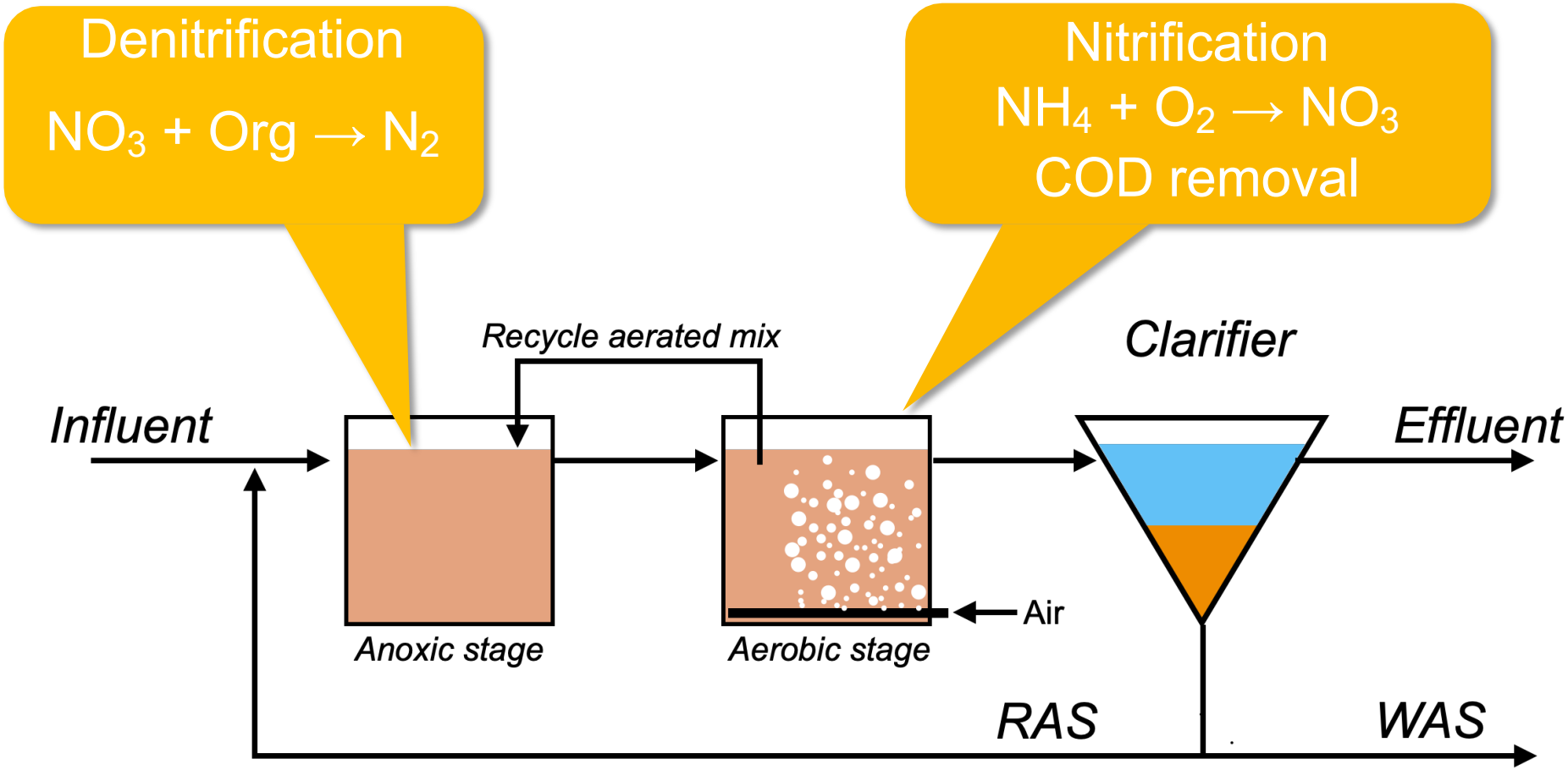
Can you identify the issues with this configuration?

Wuhrmann Process Nitrification + Denitrification



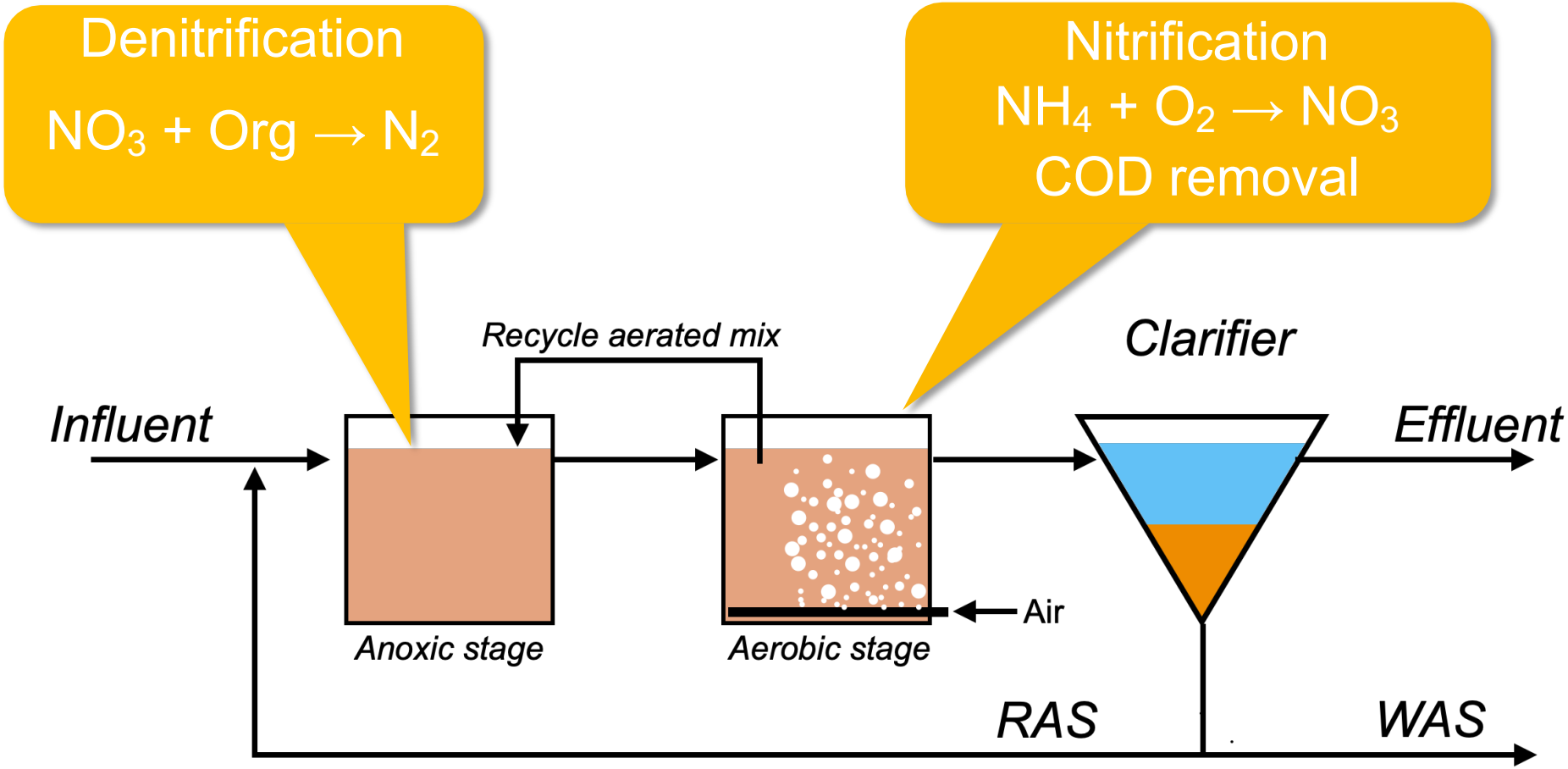
- Most of the COD is removed in the aerobic stage
- Requires COD addition (methanol) to complete denitrification

Modified Ludzack-Ettinger (MLE) Process Nitrification with Pre-Denitrification



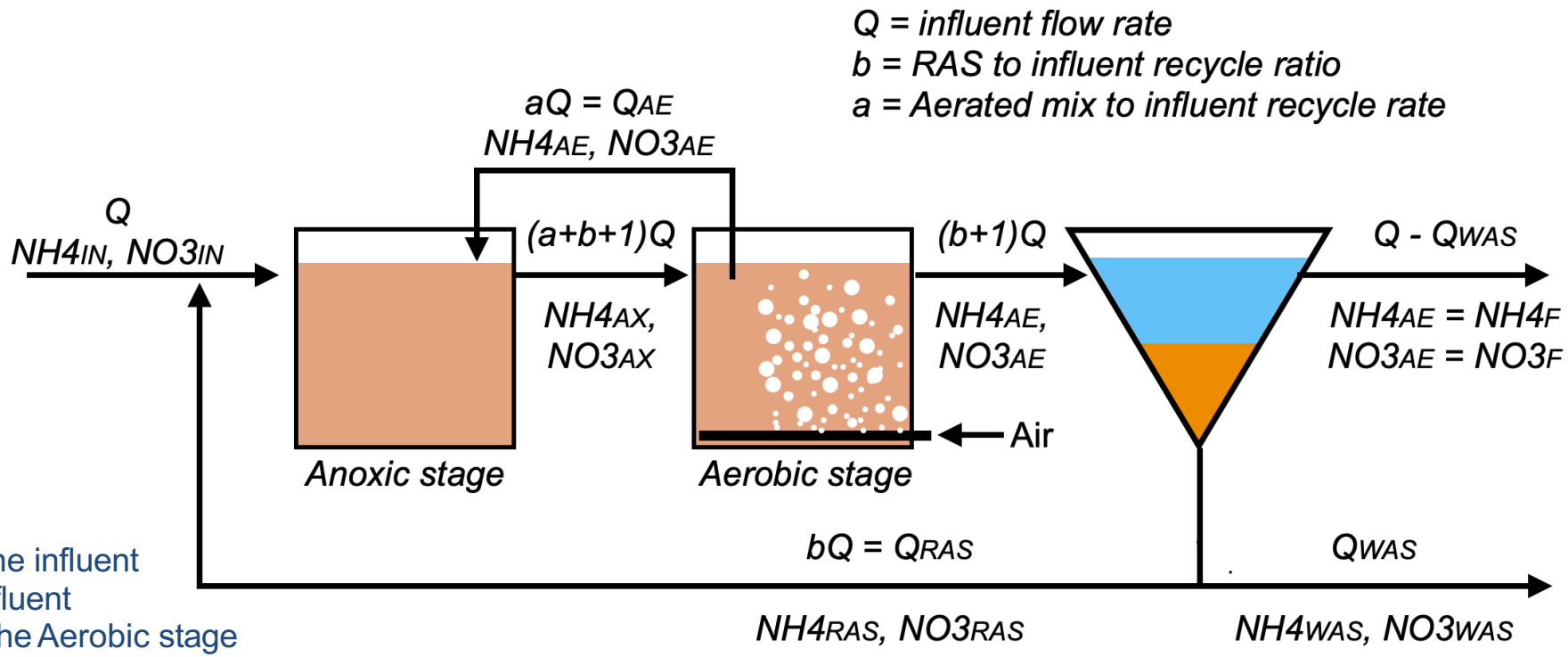
Can you identify the benefits of this configuration compared to the previous?

Modified Ludzack-Ettinger (MLE) Process Nitrification with Pre-Denitrification



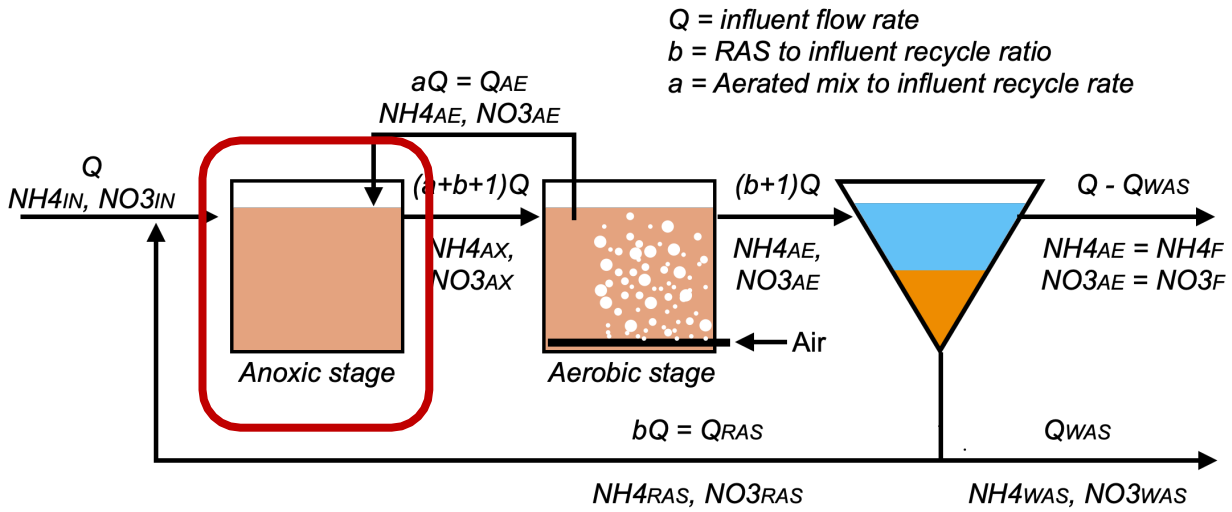
- Better use of the influent COD for denitrification (saving in additional C supply)
- Requires recycle of the aerated mix to provide NO_3 to anoxic stage

Modified Ludzack-Ettinger (MLE) Process Importance of recycling the aerated mix



$NH4_{IN}$ = Ammonium in the influent
$NO3_{IN}$ = Nitrate in the influent
$NH4_{AE}$ = Ammonium in the Aerobic stage
$NO3_{AE}$ = Nitrate in the Aerobic stage
$NH4_{AX}$ = Ammonium in the Anoxic stage
$NO3_{AX}$ = Nitrate in the Anoxic state
$NH4_{RAS}$ = Ammonium in the RAS
$NO3_{RAS}$ = Nitrate in the RAS
$NH4_{WAS}$ = Ammonium in the WAS
$NO3_{WAS}$ = Nitrate in the WAS

Appropriate design of the recycle of aerated mix is important for determining the effluent quality



- **Mass balance for NH_4 in the anoxic stage:**

$$NH4_{IN}*Q + NH4_{RAS}*b*Q + NH4_{AE}*a*Q - NH4_{AX}*(a+b+1)*Q = 0$$

To achieve complete nitrification in the aerobic stage we must have:

- $NH4_{AE} \sim 0$
- $NH4_{RAS} \sim 0$
- $NH4_{AX} = NO3_{AE} = NO3_F$

It derives that:

$$NH4_{IN}*Q + NH4_{RAS}*b*Q + NH4_{AE}*a*Q - NO3_F*(a+b+1)*Q = 0$$

As typically $b \sim 1\text{-}1.5$, a must be high to allow for low NO_3 in the treated effluent

$$NO3_{AE} = NO3_F = \frac{NH4_{IN}}{a + b + 1}$$

Biological treatment kinetics

- **Kinetics** are also important in terms of the **mass of organisms**, which may increase or decrease with time, depending on the growth conditions and **availability of substrate**.
- **Kinetics** analysis of a bioreactor involves at least **3 mass balances**, quantifying the flow and changes in**:**
  1. Water
  2. Organic matter
     - Substrate (S), measured by mg/L of BOD (or mg/L of COD)
  3. Microbial mass (biomass)
     - Biomass (X), measured by volatile suspended solids (VSS) or volatile solids (VS)

Biological treatment kinetics

- **Rate of biomass growth**

$$r_g = \frac{dX}{dt}$$

Growth rate

» r_g = growth rate of biomass, mg/L.d

» X = concentration of biomass, mg/L (typically mg VSS/L)

» t = time, d

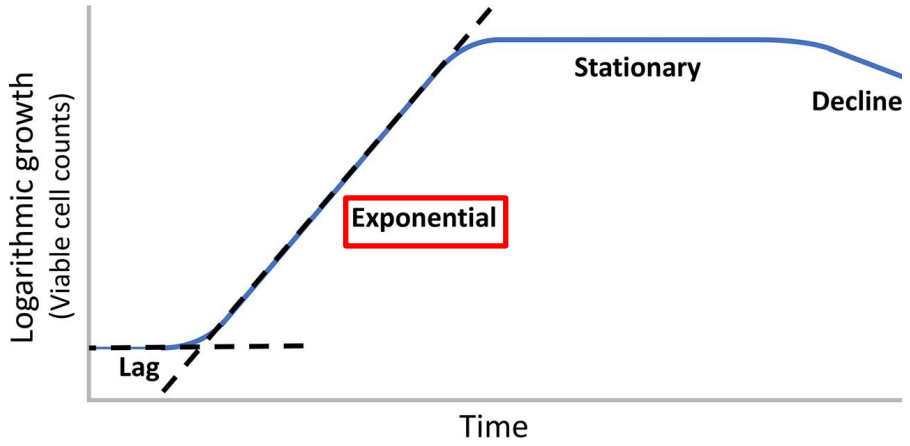
Can also be h, min

- **Rate of biomass growth (log phase)**
  - We can assume 1st order kinetics, to get:

$$r_g = \frac{dX}{dt} = \mu X$$

» μ = specific growth rate, d^{-1}

*Assumption: all microbes in the population would continue to reproduce

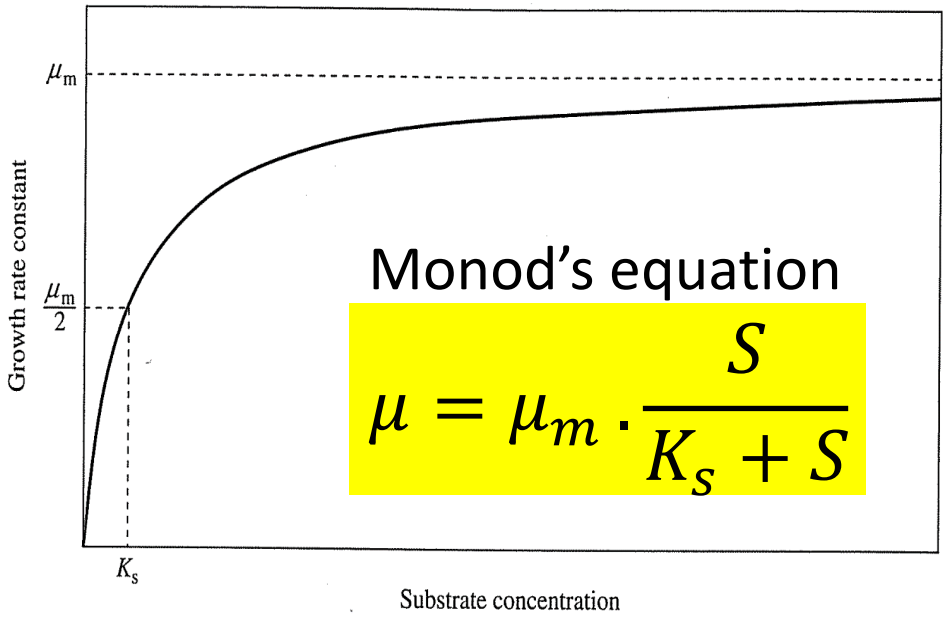


Biological treatment kinetics

- **Given that growth rate constant is dependent on substrate availability**
  - Monod (1949) reasoned that:

    - when the substrate concentration is zero, then the growth rate constant should be zero (S = 0 -> μ = 0);

    - when there is substrate in excess, the microbial growth rate should reach a maximum that is determined by the maximum speed at which microbes can reproduce (S >> K_s -> μ ≈ μ_m).



Monod's equation

$$\mu = \mu_m \cdot \frac{S}{K_s + S}$$

» μ = specific growth rate, d^{-1}

» μ_m = maximum specific growth rate, d^{-1}

» S = substrate concentration, mg/L. (typically mg BOD_5/L or mg COD/L).

» K_s = half velocity constant (substrate concentration when growth rate constant is half of its maximum value), mg/L.

Biological treatment kinetics

Scenario 1

When the substrate concentration is zero then the growth rate constant should be zero (S = 0 -> μ = 0):

$$\frac{S}{K_s + S} = 0 \qquad \mu = \mu_m \text{ x } 0 = 0$$

Scenario 2

When substrate concentration is in excess, the microbial growth rate should reach a maximum that is determined by the maximum speed at which microbes can reproduce (S >> K_s -> μ ≈ μ_m):

$$\frac{S}{K_s + S} \begin{matrix} \longrightarrow \infty \\ \longrightarrow \infty \end{matrix} = \frac{\infty}{\infty} = 1 \qquad \mu = \mu_m \text{ x } 1 \approx \mu_m$$

Scenario 3

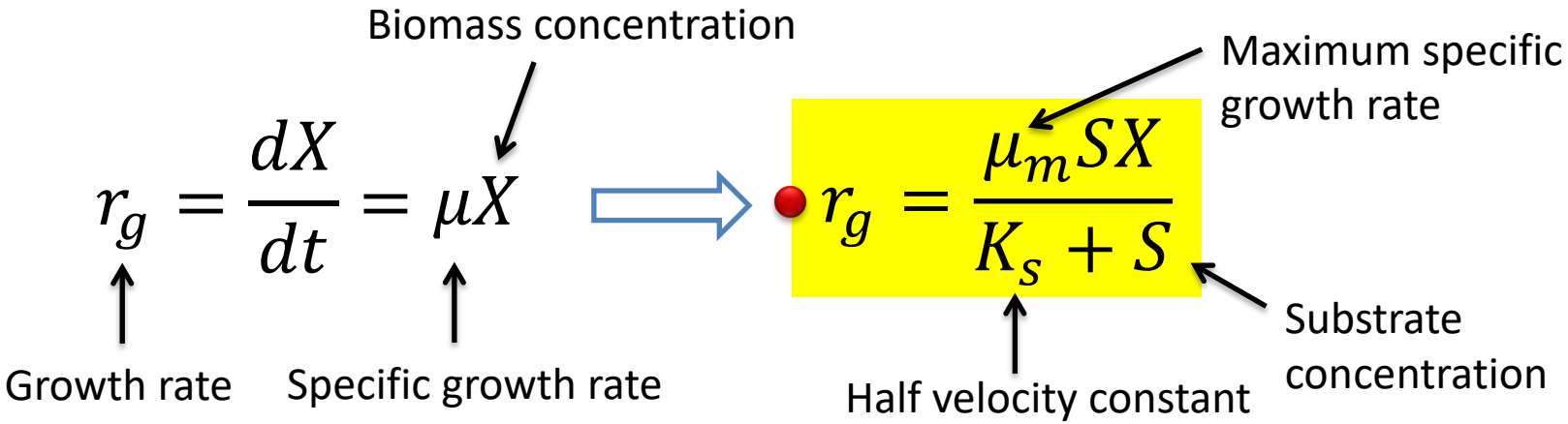
At low substrate concentration (S << K_s):

$$\frac{S}{K_s + S} \approx \frac{S}{K_s} \qquad \mu = \mu_m \cdot \frac{S}{K_s}$$

Biological treatment kinetics

- Linking biomass growth rate to substrate availability through Monod Equation

Monod Equation: $$\mu = \mu_m \cdot \frac{S}{K_s + S}$$



*Assumption: all microbes in the population would continue to reproduce

Biological treatment kinetics

- Rate of biomass growth increases with a corresponding decrease in rate of substrate utilization
  - Amount of microbial mass generated should be proportional to amount of substrate consumed

$$\frac{dX}{dt} \propto -\frac{dS}{dt}$$

  - A proportion of substrate consumed will be converted to new microbial mass.

$$\frac{dX}{dt} = Y\left(-\frac{dS}{dt}\right)$$

$$r_g = Y(-r_{su})$$

» Y = biomass yield

» r_{su} = substrate utilization rate, mg/L.d

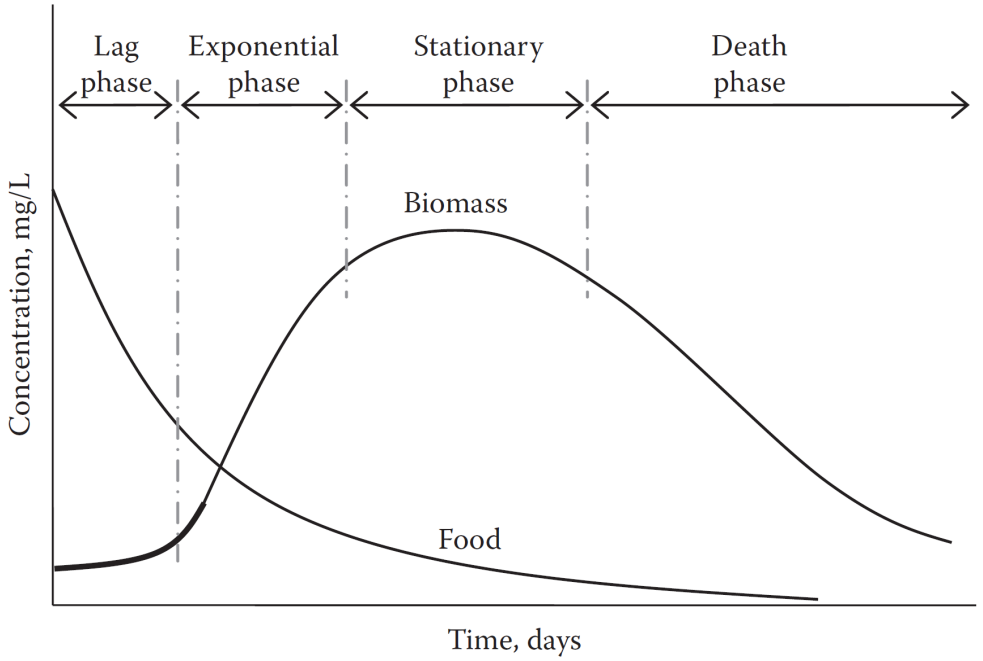


Figure 8.1 Relationship between microbial growth and substrate utilization.

Biological treatment kinetics

$$Y = \frac{mg\ biomass\ produced}{mg\ substrate\ utilized} = \frac{mg\ VSS\ produced}{mg\ BOD\ removed}$$

- The maximum specific growth rate, is related by yield coefficient to the maximum specific substrate utilization rate, *k*:

$$Y = \frac{r_g}{-r_{su}} \quad \text{or} \quad Y = \frac{\mu_m}{k}$$

μ_m: Maximum specific substrate utilization rate

$$r_g = \frac{\mu_m S X}{K_s + S} \Rightarrow r_{su} = \frac{-kSX}{K_s + S}$$

Biological treatment kinetics

- Since rate of biomass growth, r_g, is approx. 1st order kinetics,
- Rate of biomass death, r_d, should also be a 1st order kinetics but with a negative term:

Growth

$$r_g = \frac{\mu_m S X}{K_s + S}$$

Death

$$r_d = -k_d X$$

» k_d = endogenous decay rate constant, time^{-1}

» r_d = death rate of biomass, mg/L.d

- The net biomass growth, r'_g :

$$r'_g = r_g + r_d = \frac{\mu_m S X}{K_s + S} - k_d X$$

or $$r'_g = -Y r_{su} - k_d X$$

Biological treatment kinetics

TABLE 7

Typical Microbial Kinetic Parameters (All are given for 20°C)

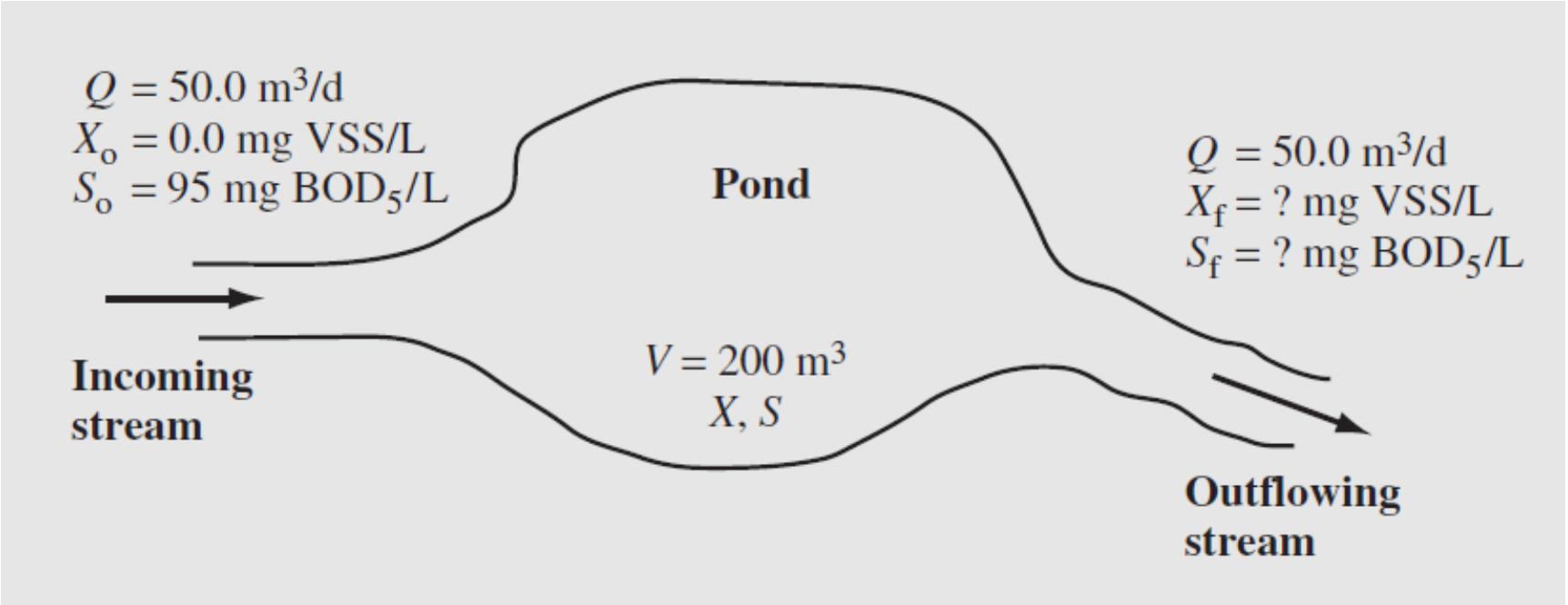
Parameter	Typical Units	Range	Default Value
k	mg BOD_5/(mg VSS · d)	2–10	5
μ_m	d^{-1}	1–10	3
K_s	mg BOD_5/L	25–100	60
Y	mg VSS/mg BOD_5	0.2–0.8	0.6
k_d	d^{-1}	0.025–0.075	0.06

Source: Adapted from Metcalf and Eddy, 1991.

Example 6.9 BOD consumption in pond

The shallow pond depicted in the figure stays well mixed due to the wind and the steady flow through of a small creek. If the microbes in the pond consume the inflowing biodegradable organic matter according to typical kinetics, determine:

a. The BOD5 leaving in the pond
b. The biodegradable organic matter removal efficiency of the pond
c. The concentration of VSS leaving the pond



Note: microbial kinetic parameters refer to Table 7

Example 6.9 BOD consumption in pond

Solution:

a. Mass balance on the microbial mass using the pond as control volume.

$$0 = QX_0 - QX_f + Vr_g' \,, \quad r_g' = \frac{\mu_m SX}{K_s + S} - k_d X$$

recognizing that $X_0 = 0$, $X_f = X$, and $S_f = S$,

$$0 = -QX + V\frac{\mu_m SX}{K_s + S} - Vk_d X$$

Because the HRT is $\theta = V/Q$, rearrange the above equation as:

$$1 = \theta\frac{\mu_m SX}{K_s + S} - \theta k_d$$

$$S = \frac{K_s(1 + \theta k_d)}{\theta\mu_m - 1 - \theta k_d}$$

Using the default values in Table 7 and $\theta = V/Q = 200\ m^3/50\ m^3/d = 4$ d, the outflowing BOD_5 concentration is:

$$S = \frac{60\times(1 + 4\times0.06)}{4\times3 - 1 - 4\times0.06} = \mathbf{6.9\ mg\ BOD_5/L}$$

b. The efficiency of the pond is:

$$\text{Eff} = \frac{S_0 - S_f}{S_0}\times100\% = \frac{95 - 6.9}{95}\times100\% = \mathbf{93}\%$$

Example 6.9 BOD consumption in pond

Solution:

c. To calculate the microbe concentration (as VSS) in the outflowing stream, we construct a steady state mass balance for the pond as:

$$0 = QS_0 - QS_f + Vr_{su}, \qquad r_{su} = \frac{-kSX}{K_s + S}$$

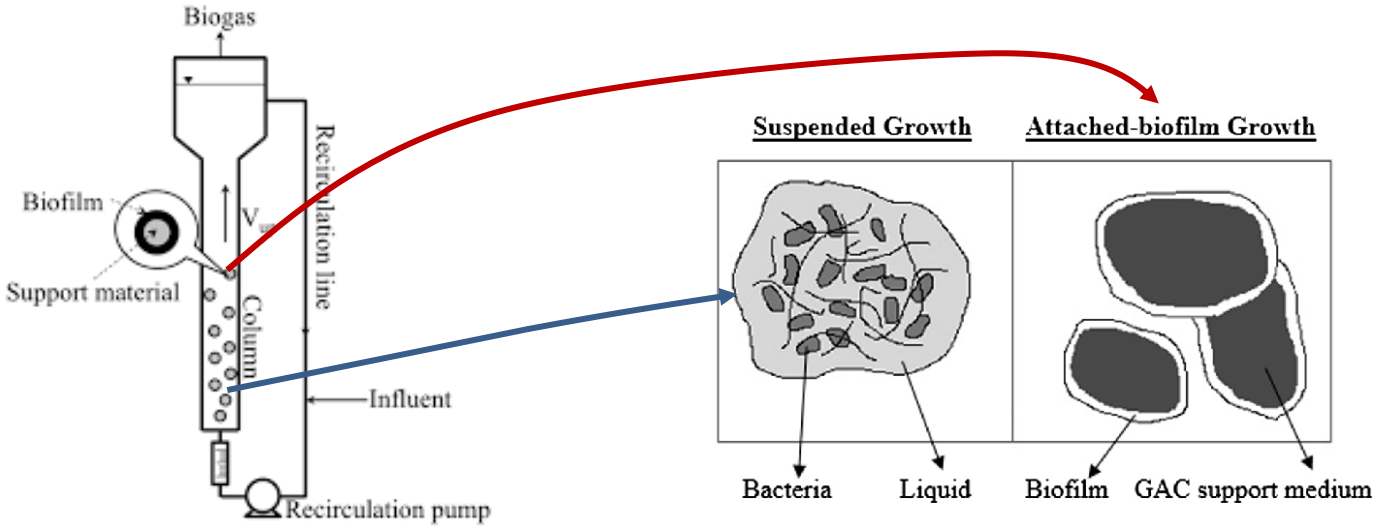
$$X = \frac{(K_s + S)(S_0 - S)}{\theta k S}$$

Again, using the default values in Table 7

$$X = \frac{(60 + 6.9)\times(95 - 6.9)}{4\times5\times6.9} = \mathbf{43\ mg\ VSS/L}$$

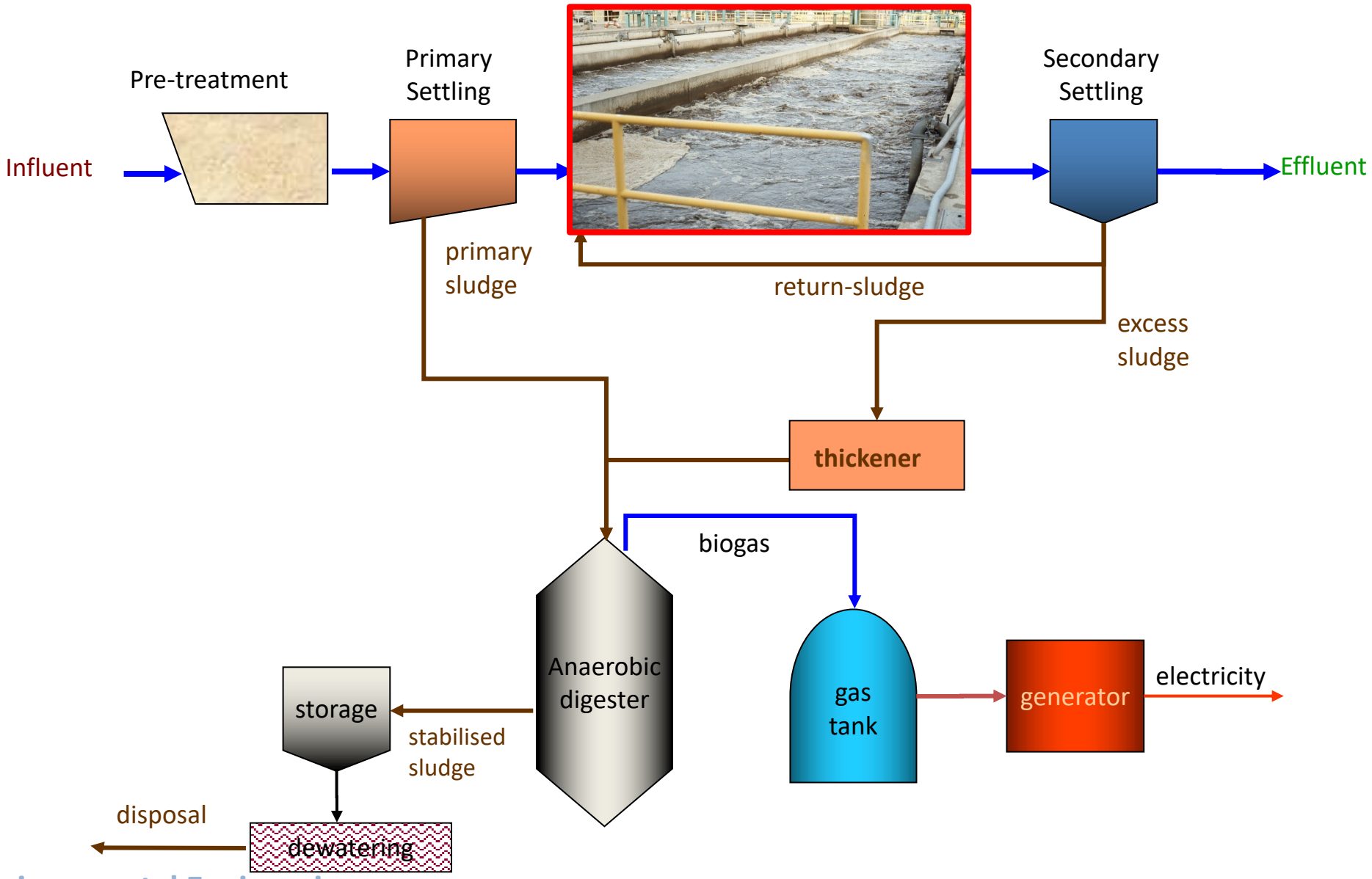
Growth of microorganisms

- Suspended-growth
  - Activated sludge processes
  - Oxidation ponds and aerated lagoons
- Attached-growth (fixed-film or biofilm)
  - Trickling filters
  - Rotating biological contactor
- Hybrid suspended/attached growth systems



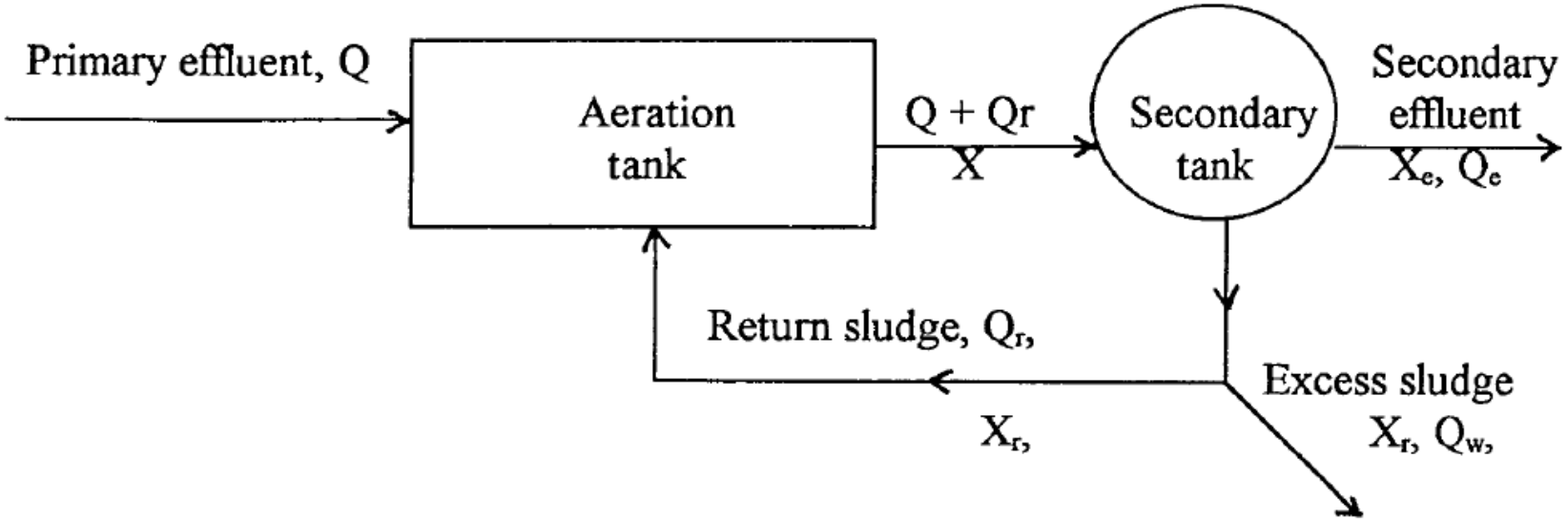
(source: McCarty et al., 2015, Anaerobic Biotechnology, pp. 211-242; Jamali et al., 2017, Energ. Convers. and Manage., 141, pp. 354-366)

Activated sludge process (ASP)



Activated sludge process (ASP)

- Key features
  - Oxygen provided (aeration)
  - Returning of sludge
  - The term "**Activated"** is referring to the return of **active** biomass from the secondary clarifier to the aeration basin.



Activated sludge process (ASP)

Design parameters for ASP

- Food To Micro-organism Ratio (F/M)
- Hydraulic Retention Time (HRT, q)
- Mean Cell Residence Time (or Solids Retention Time, SRT or Sludge Age)

F/M ratio

- An important design and operation parameter indicating the relationship between the **organic loading** and the **amount of micro-organisms** in the activated sludge aeration tank.
- Synonymous Terms To Be Noted
  - The **F** of the F/M ratio refers to **Food**: influent, substrate (S), feedstock, COD, BOD, organics or organic wastes.
  - The **M** of the F/M refers to **micro-organism**: microbes, biological solids, biomass (X), microbial cells, volatile suspended solids (VSS), volatile solids (VS).
  - Aeration tank, aeration chamber, aeration basin, reactor, or bioreactor.

Activated sludge process (ASP)

The food is expressed as kg of BOD supplied to the aeration tank per day.

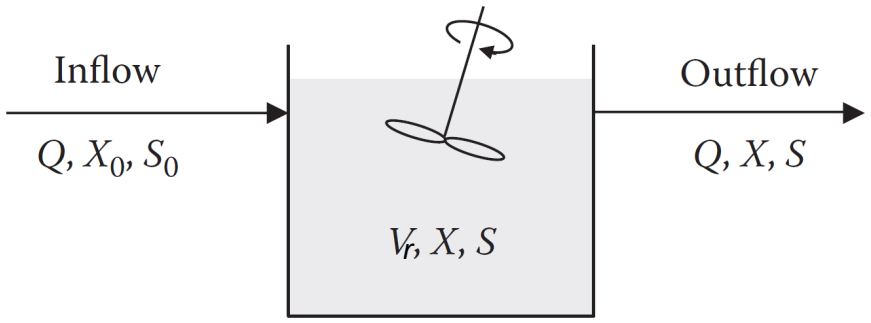
$$\mathbf{F = Q \cdot S_o} \text{ (g BOD/d)}$$

where Q = influent flow rate, m^3/d, S_o = influent BOD, mg/L.

The micro-organisms are expressed as kg of volatile suspended solids of the 'mixed liquor' in the tank.

$$\mathbf{M = V_r \cdot X} \text{ (g VSS)}$$

where V_r = volume of aeration tank, m3, X = micro-organism concentration, mg/L.



$$\frac{F}{M} = \frac{Q \cdot S_O}{V_r \cdot X}$$

Activated sludge process (ASP)

Mean cell residence time

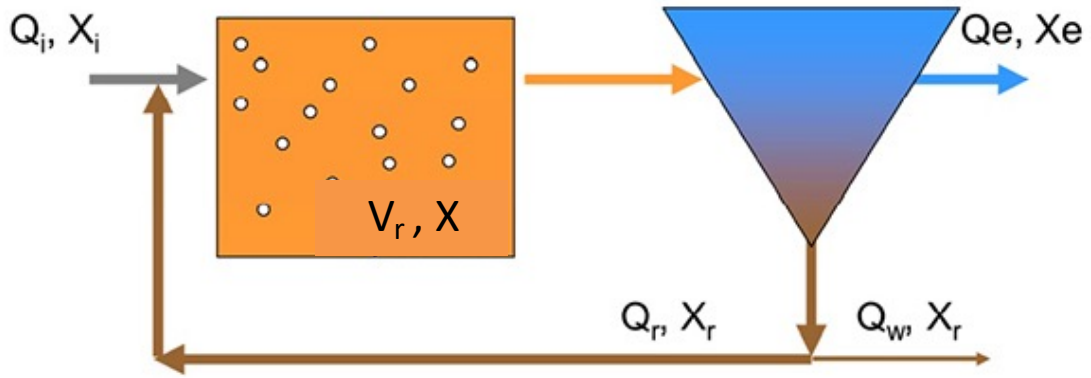
- Also termed sludge age or solids retention time (SRT)
  - the average **time** the **sludge solids** stay in the system

$$\theta_c = \frac{V_r X}{Q_e X_e + Q_w X_r}$$

If $X_e \approx 0$

$$\theta_c \approx \frac{V_r X}{Q_w X_r}$$

θ_c = *mean cell residence time, d*
X = microbial concentration in the aeration tank, mg/L
X_e = *microbial concentration in the secondary effluent, mg/L*
X_r = *microbial concentration in the waste sludge, mg/L*
Q_e = *flowrate of effluent wastewater, m^3/d*
Q_w = *flowrate of waste sludge, m^3/d*



Oxidation ponds

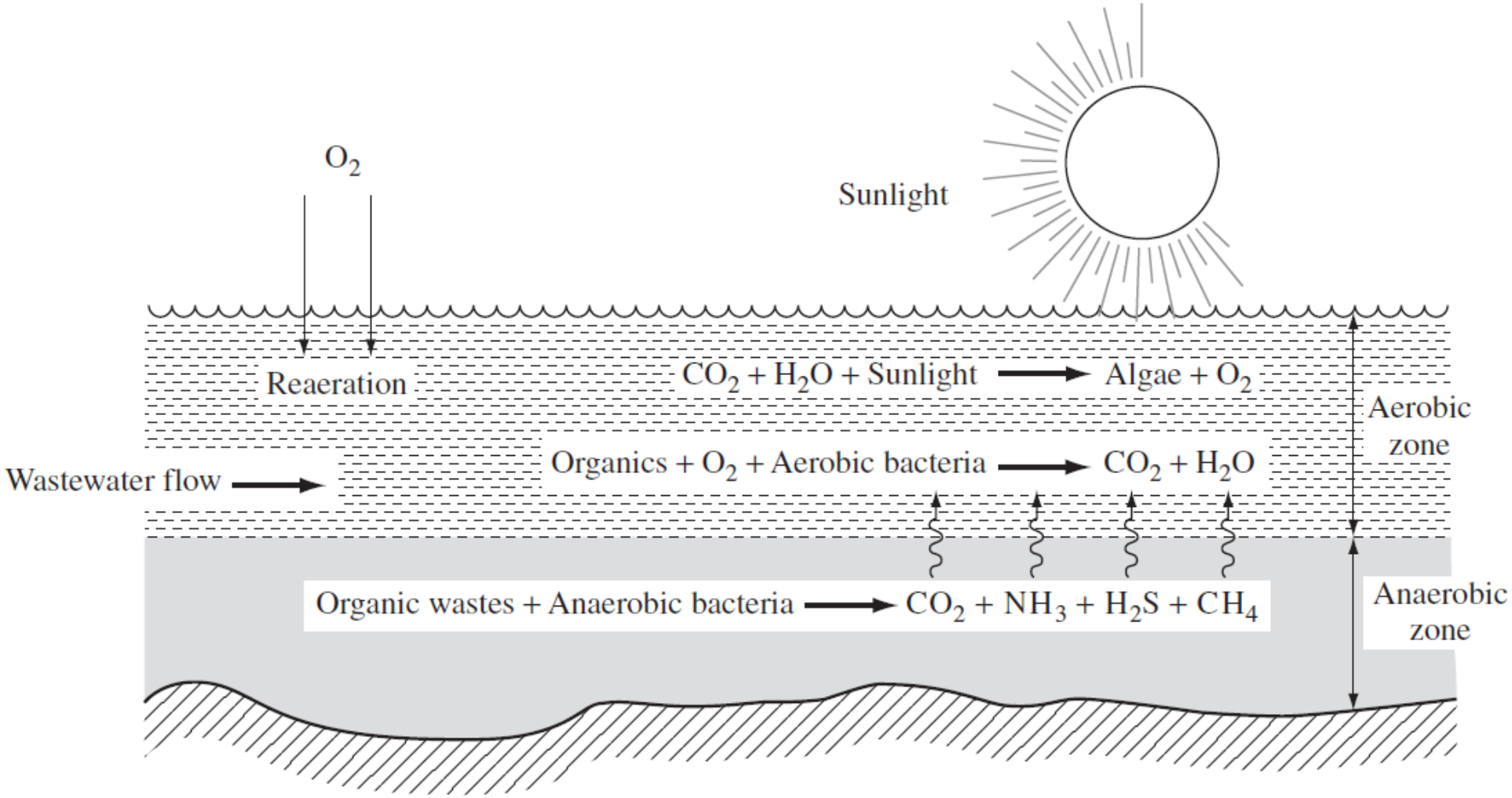
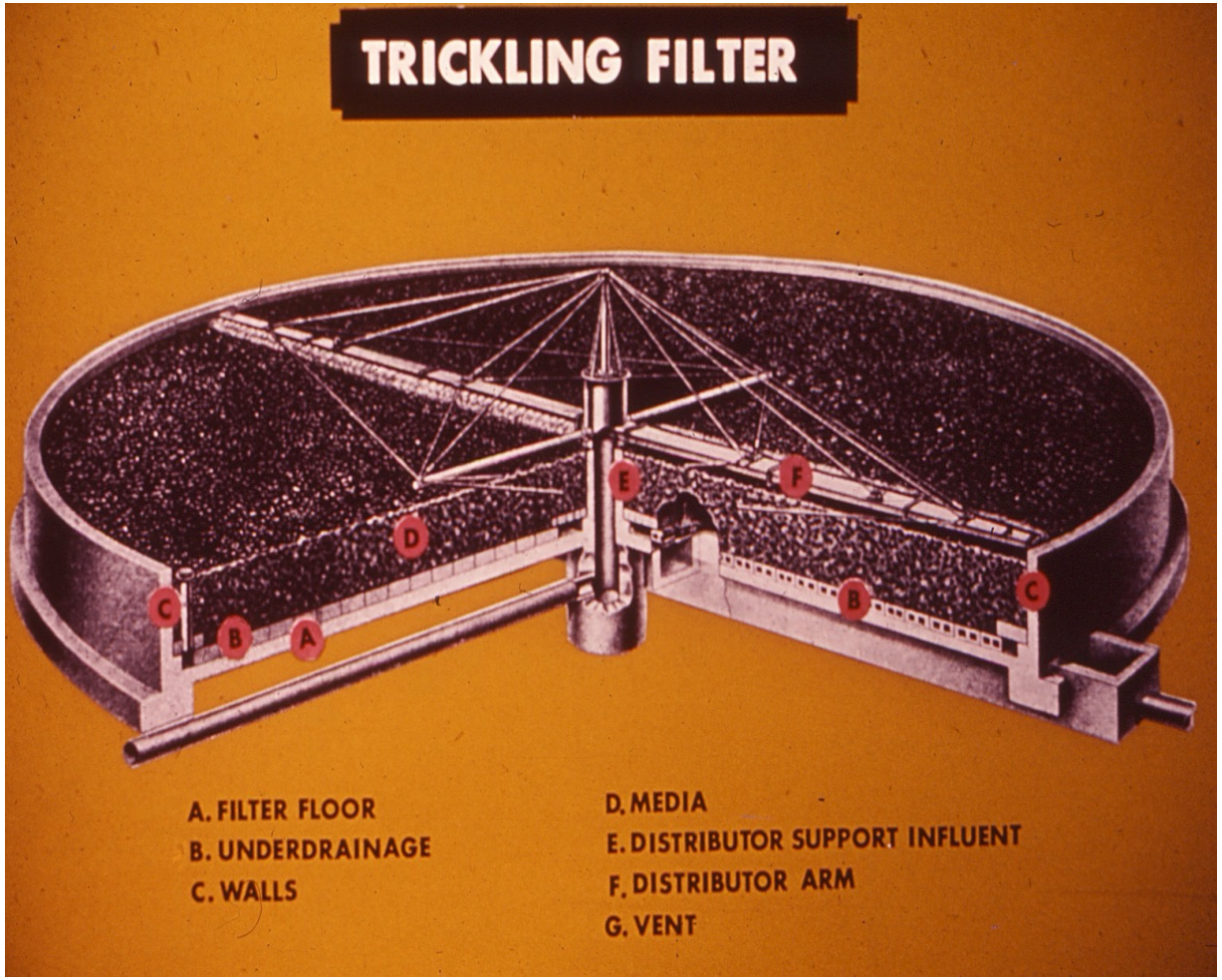
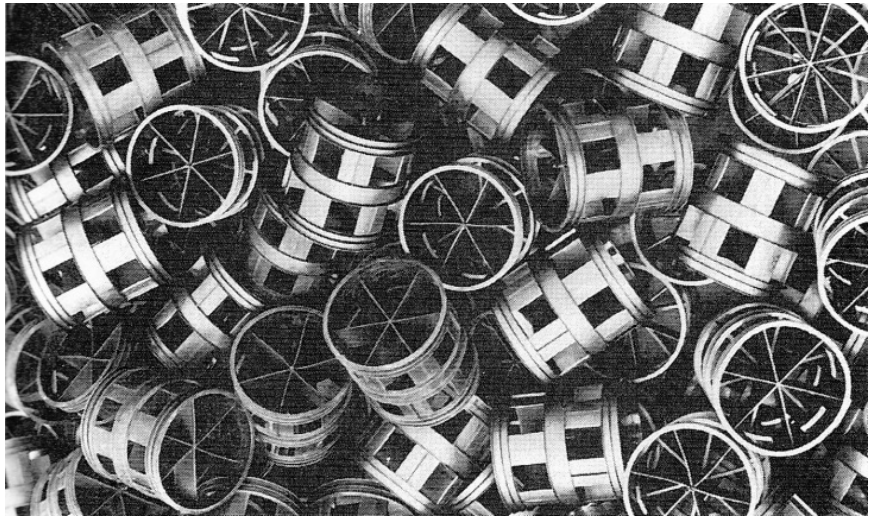


FIGURE 17 Schematic of an oxidation pond.

Trickling filters



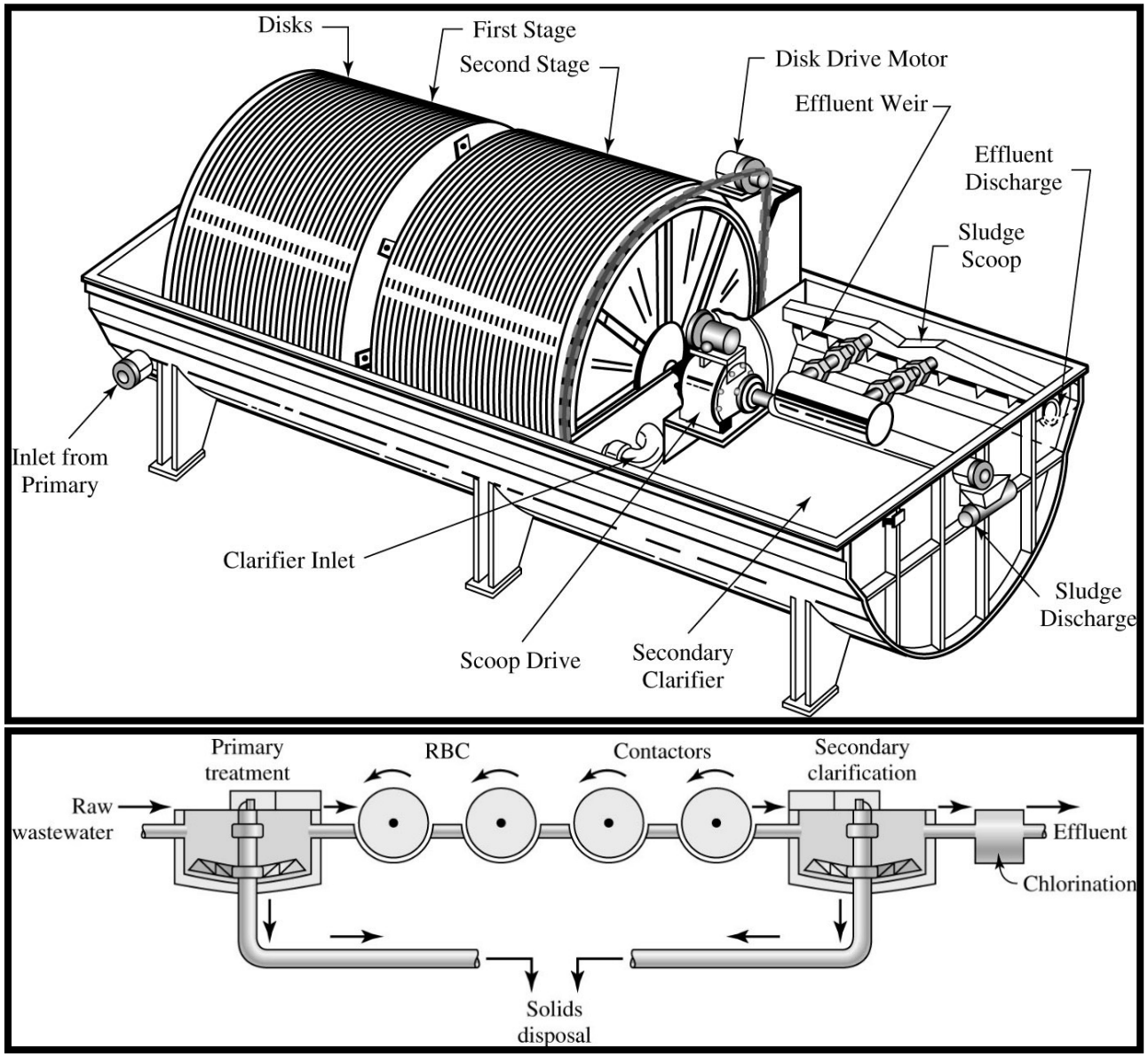
Davis and Cornwell 2008



Source: Viessman and Hammer



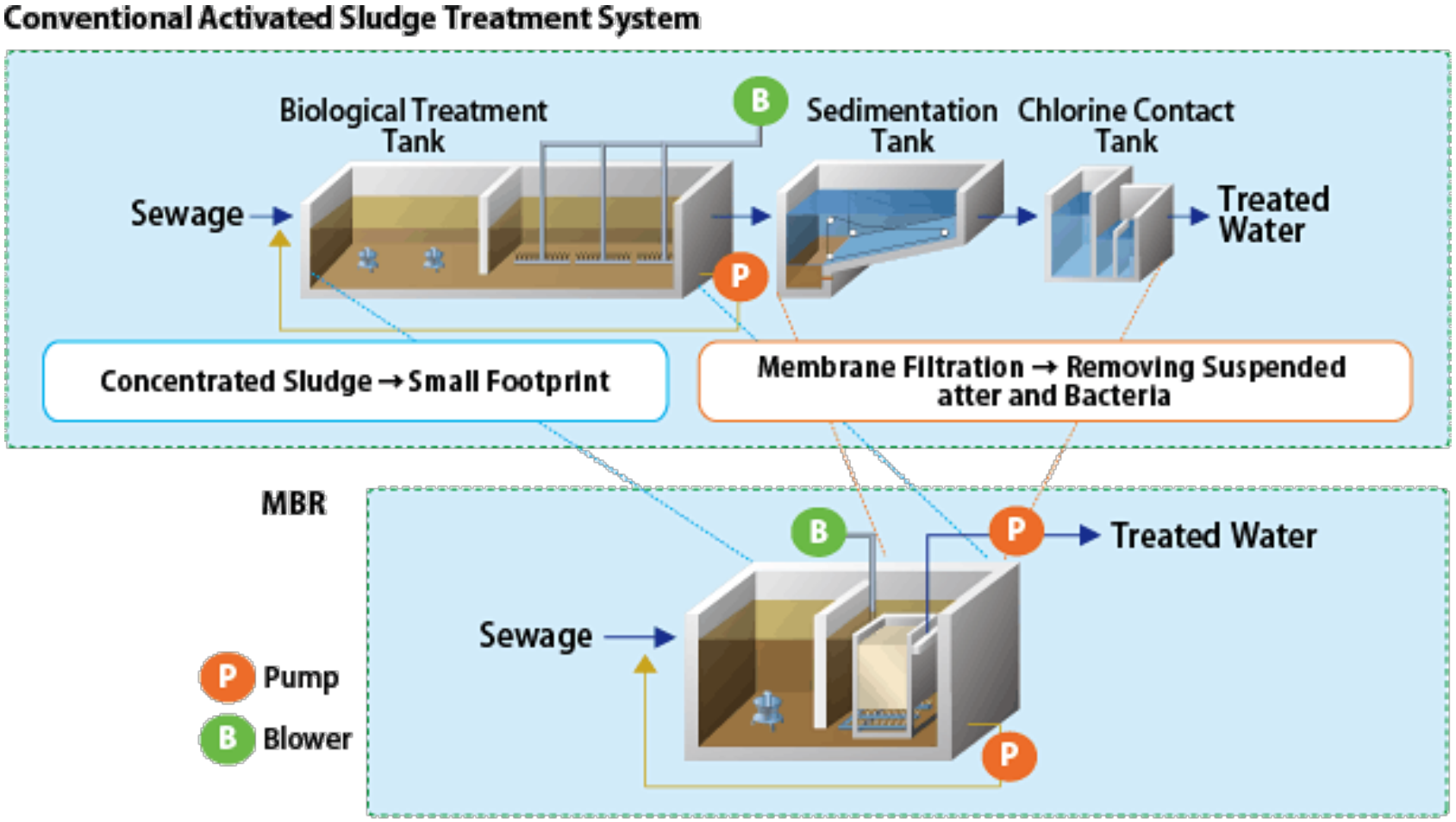
Rotating biological contactors (RBC)



Davis and Cornwell 2008

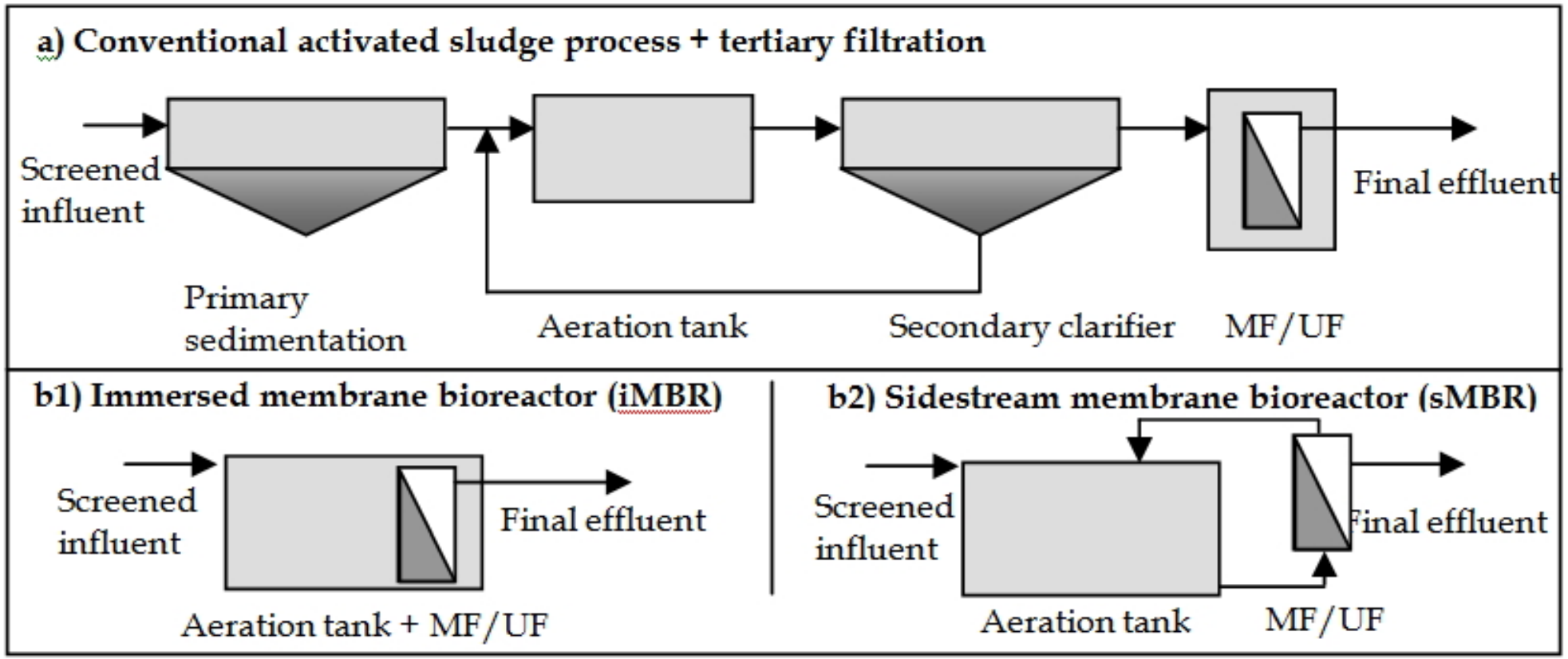
Membrane Bioreactor (MBR)

- Removal of particulate solids (producing low TSS effluent)
- Produces high-quality permeate



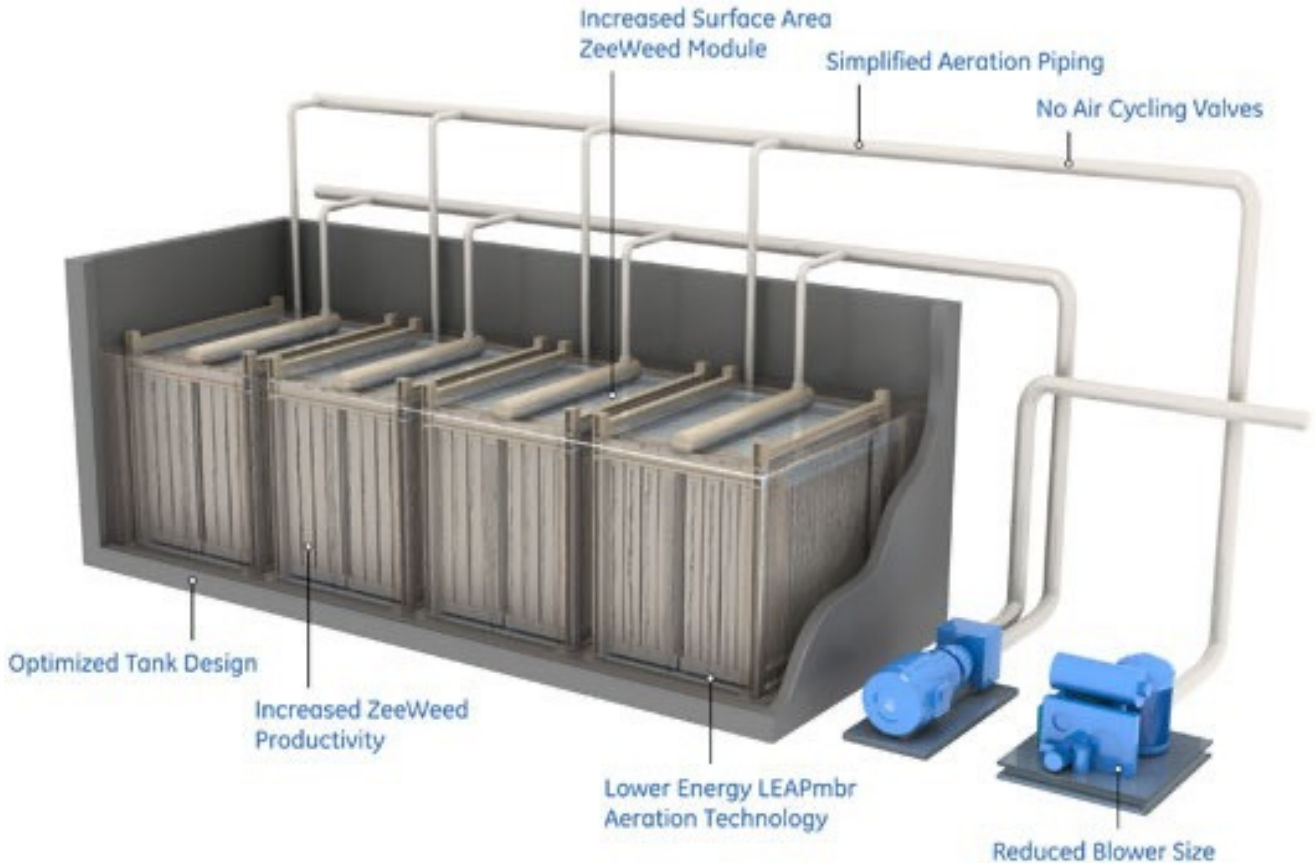
Package Sewage Treatment Plants, Rami E Kremesti M.

Membrane bioreactors (MBR)



Membrane Bioreactor (MBR)

A combination of a suspended growth biological treatment method, usually activated sludge, with membrane filtration equipment, typically low-pressure microfiltration membrane.



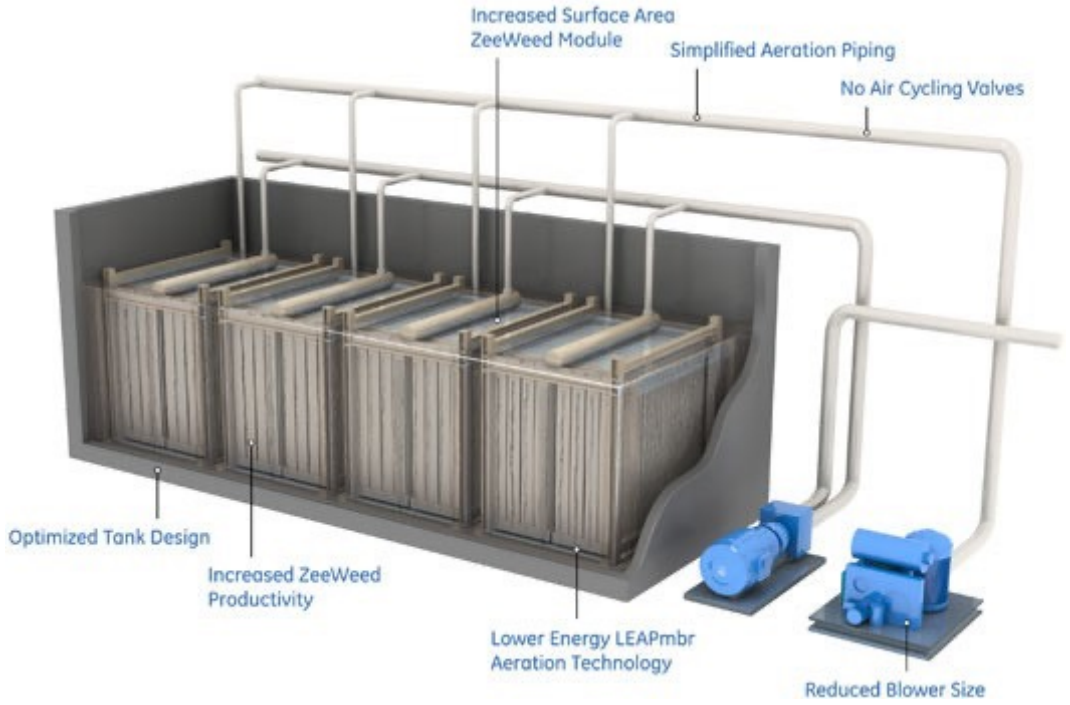
https://www.amtaorg.com/Membrane_Bioreactors_for_Wastewater_Treatment.html

Membrane Bioreactor (MBR)

Biological reactor and membrane filtration as a integrate system for the secondary treatment of wastewater

Membranes perform the solid/liquid separation of the final effluent through filtration

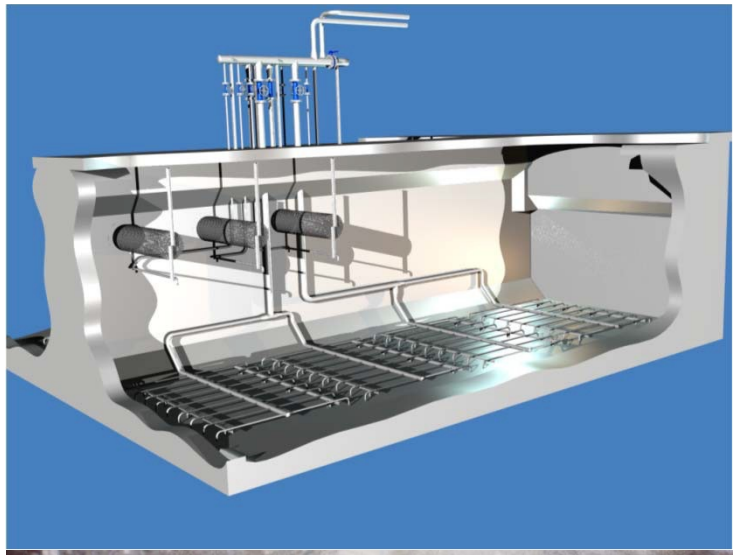
Filtration takes place by the application of a vacuum pressure

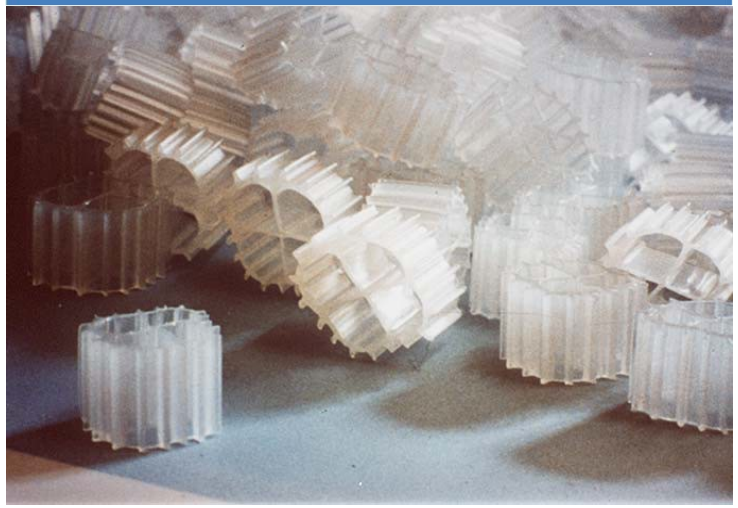


The Moving Bed Biofilm Reactor (MBBR)



The components of the MBBR treatment system





THE MAJOR COMPONENTS

a) Tank
b) Media
c) Aeration System
d) Sieve Assemblies
e) Blowers
f) Mixers

Carrier filling fraction (%)
(bulk volume occupied by media in empty reactor):

Possible filling fraction: 0 – 70 %
Normal filling fraction: 50 – 65 %

MBBR carriers



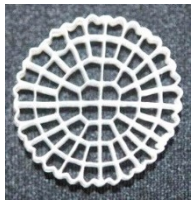
- 500 m^2/m^3 bulk
- 9.1 x 7.2 mm diameter/depth

K1



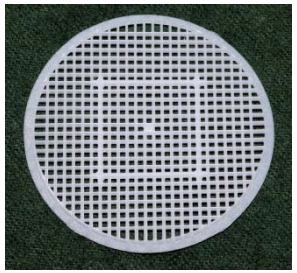
- 500 m^2/m^3 bulk
- 25 x 10 mm diameter/depth

K3



- 800 m^2/m^3 bulk
- 25 x 3.5 mm diameter/depth

K5



- 1200 m^2/m^3 bulk
- 48 x 2.2 mm diameter/depth

BiofilmChip M



Z-MBBR

Courtesy AnoxKaldnes



- 650 m^2/m^3 bulk
- 13 x 13 mm diameter/depth

ABC

Courtesy Aqwise



- 650 m^2/m^3 bulk
- 18.5 x 14.5 x 7.3 mm length/width/depth

BWT S

Courtesy Biowater Technology



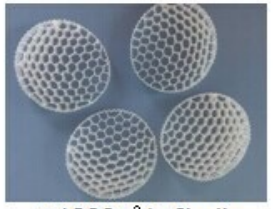
●620 m^2/m^3bulk
●25 ×10mm
Diameter/depth

SPR-2



●800 m^2/m^3bulk
●25 ×4mm
Diameter/depth

SPR-3



●1200 m^2/m^3bulk
●36 ×4mm
Diameter/depth

SPR-4

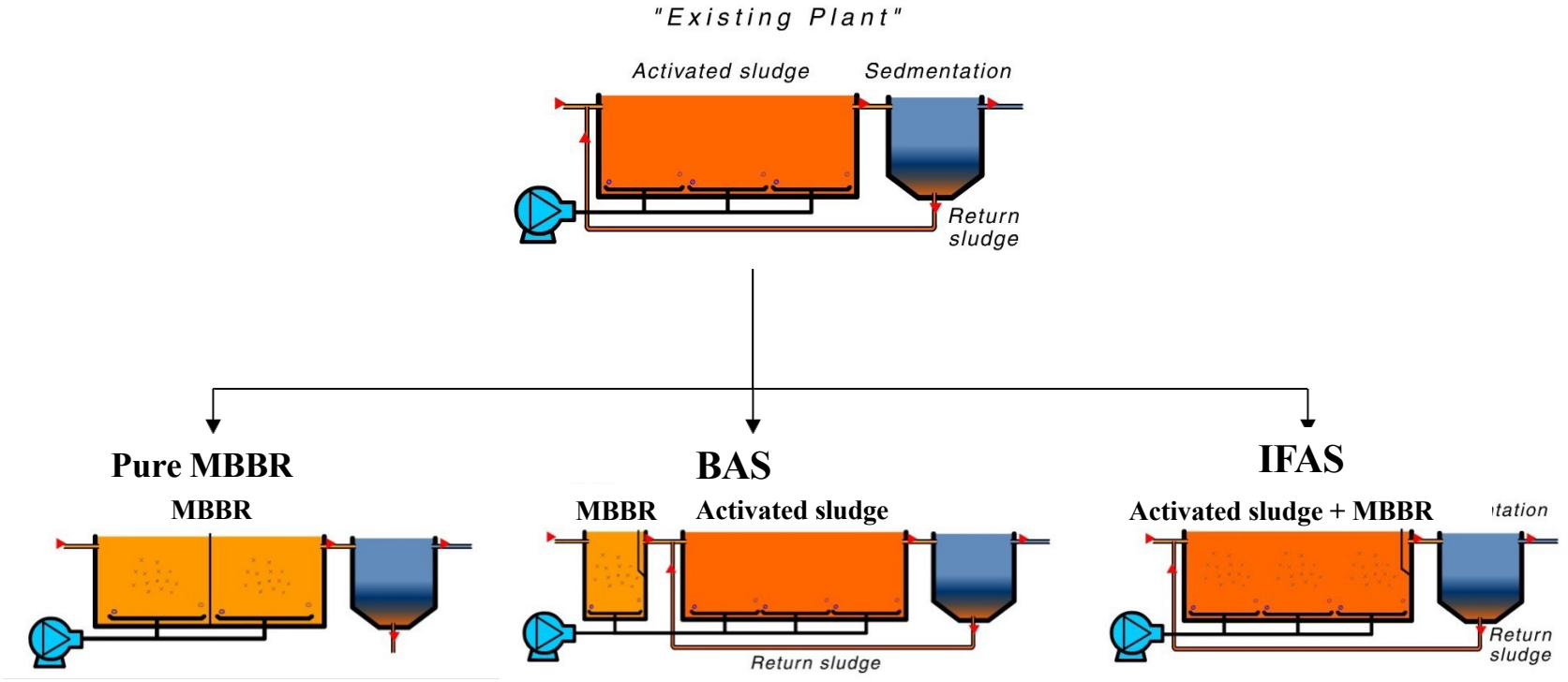
Courtesy Qingdao Spring

Carriers from the MBBR-based SANI pilot at Shatin

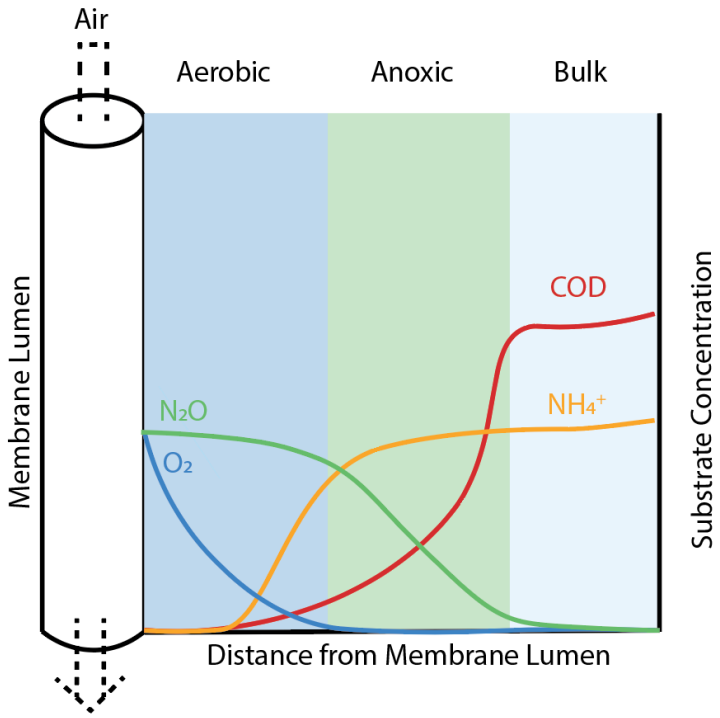


Project led by Prof GH Chen at HKUST

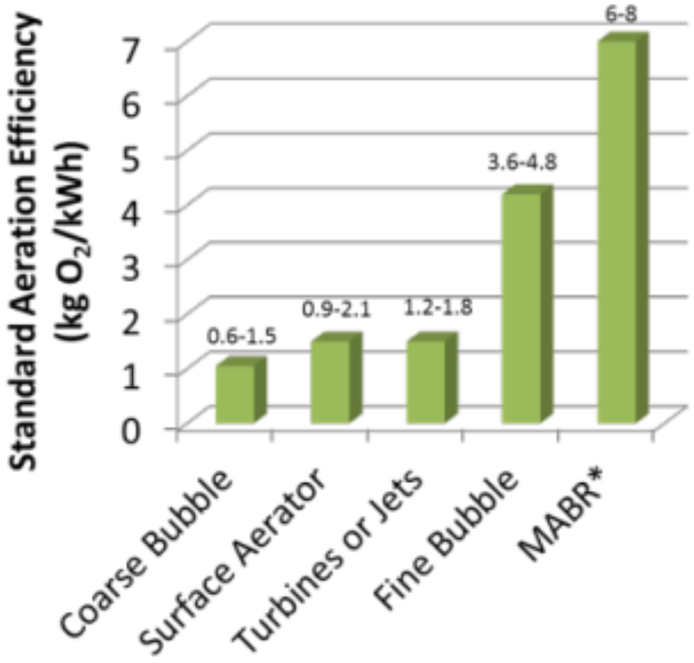
Upgrading AS-plants by the use of MBBR Three options for nitrification



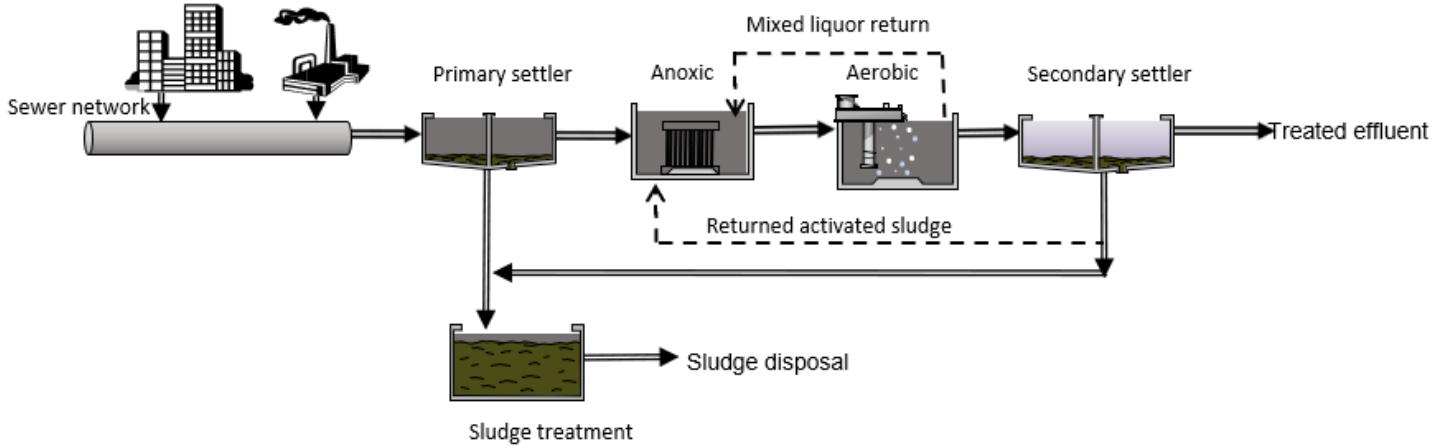
Membrane Aerated Biofilm Reactor (MABR）



- Higher OTE
- Process intensification
- Reduce internal recycle pumping
- Potential low-carbon footprint technology

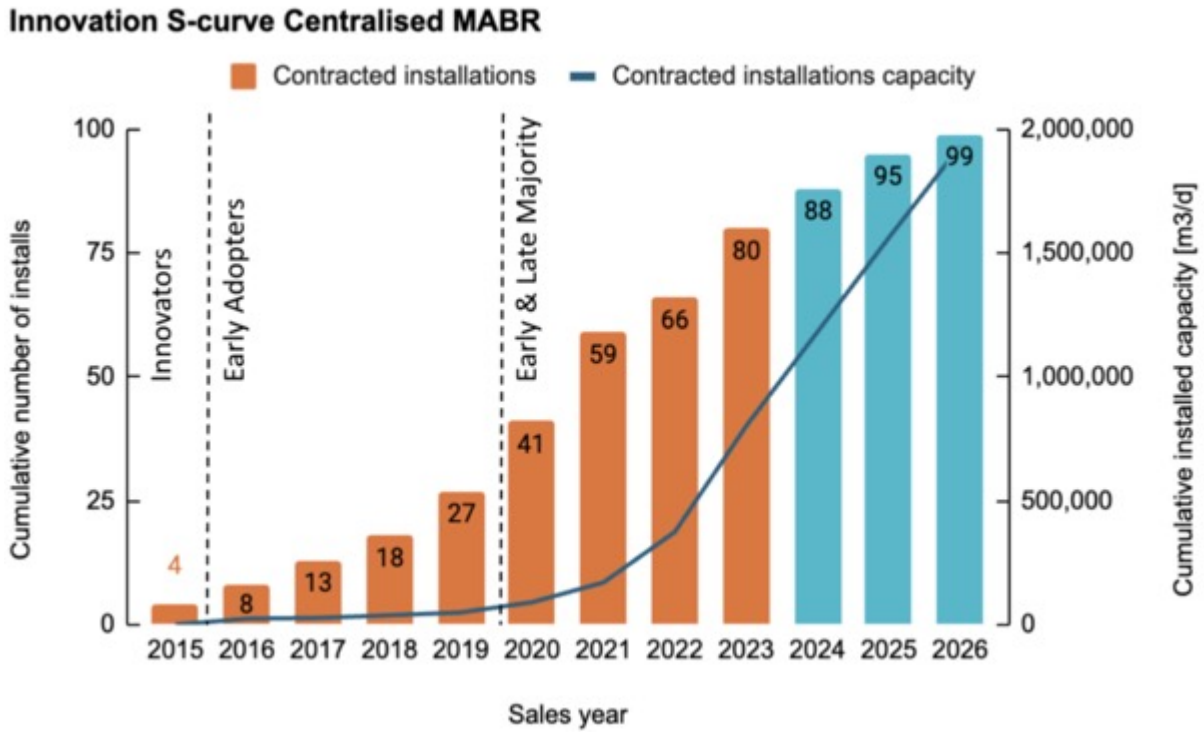


Hybrid MABR



Membrane Aerated Biofilm Reactor (MABR）

Three main suppliers currently dominate the commercial MABR market



Adoption S-curve for full-scale membrane aerated biofilm reactor (MABR) installations installed and contracted as of April, 2024 （He et al., 2025）

SUEZ （Veolia）– ZeeLung™ MABR

A gas-permeable sheet-type MABR media designed for easy retrofit in existing tank



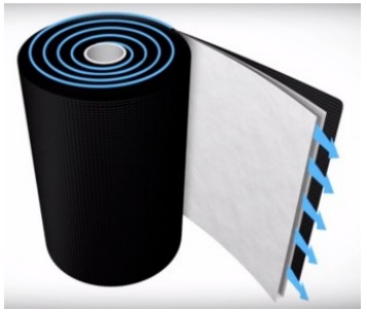
DuPont / OxyMem – OxyFAS® MABR

Hollow-fibre MABR modules for energy-efficient nitrification and plant upgrades.



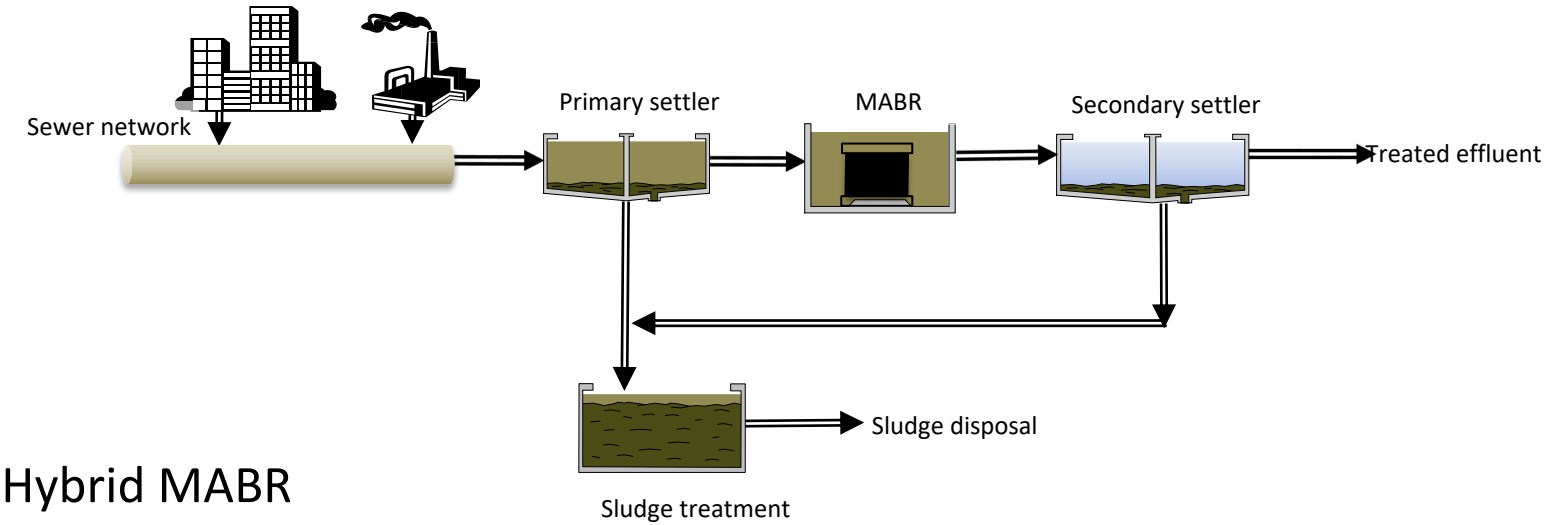
Fluence – Aspiral™ / Smart Pack MABR

Spiral-wound packaged MABR units for small and decentralised wastewater plants.

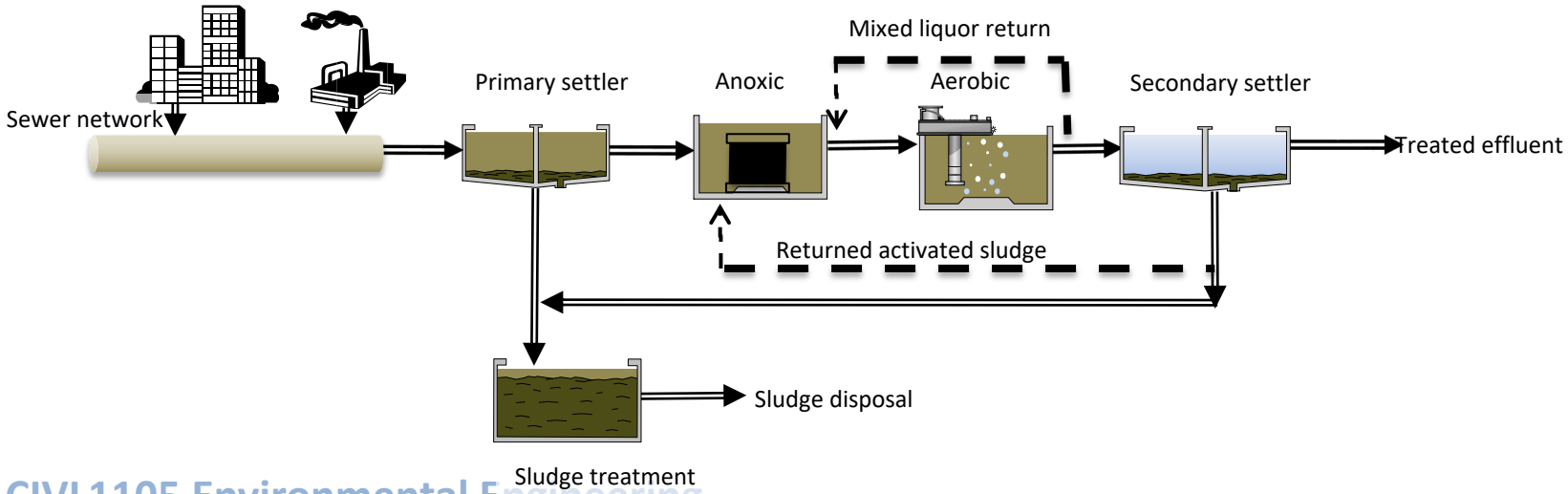


Application of MABR Technology

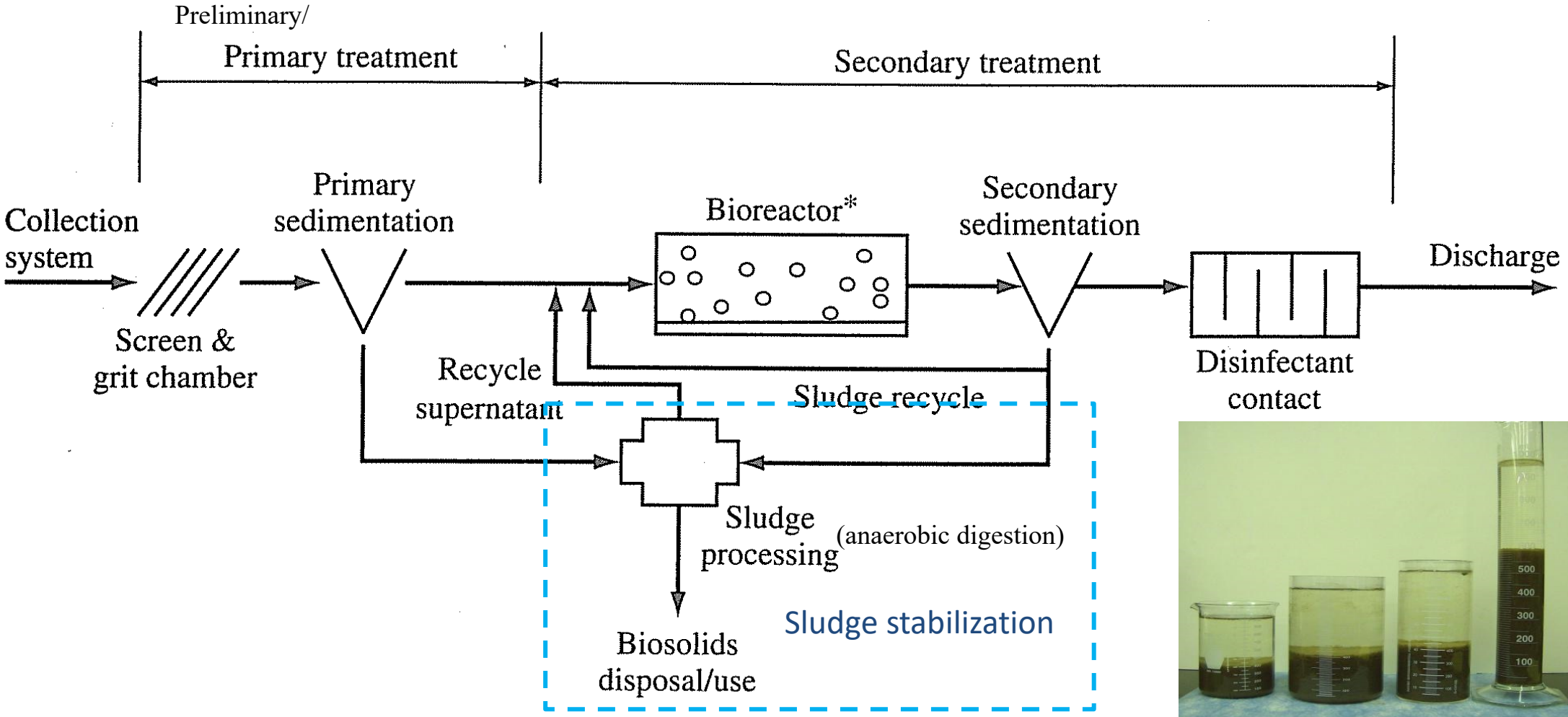
Pure MABR

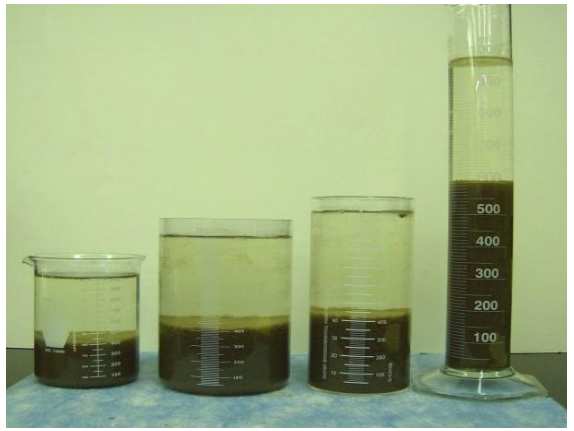


Hybrid MABR



Sludge treatment





Contains significant amount of organic matter!

Sludge treatment

Anaerobic digestion

- Biological conversion of organics to **methane and carbon dioxide** in an **oxygen-free/anaerobic environment**.
- Main advantages over aerobic processes:
  - A high degree of **waste stabilization** is possible at **high organic loads**
  - There is **low production of waste biological sludge**
  - **Less energy** is required
  - There are **low nutrient requirements**
  - Methane gas produced is a **useful source of fuel**
  - **Smaller reactor volume** is required
  - No oxygen is required, so **treatment rates are not limited by oxygen transfer rates**; **energy-saving**
  - **Rapid reactivation of biomass** is possible with substrate addition, after long periods of starvation

Sludge treatment

Two-stage anaerobic digestion

- Fermentative and acetogenic bacteria: **more tolerant** to changes in temperature and pH, and they **grow much faster**
- Acid formation → pH ↓ (pH < 7)
- Methanogen: **slow growth**, **sensitive** to temperature, pH (optimal working pH ~7), toxins and oxygen
- Two-stage AD can achieve **spatial separation of acid formers and methane formers**

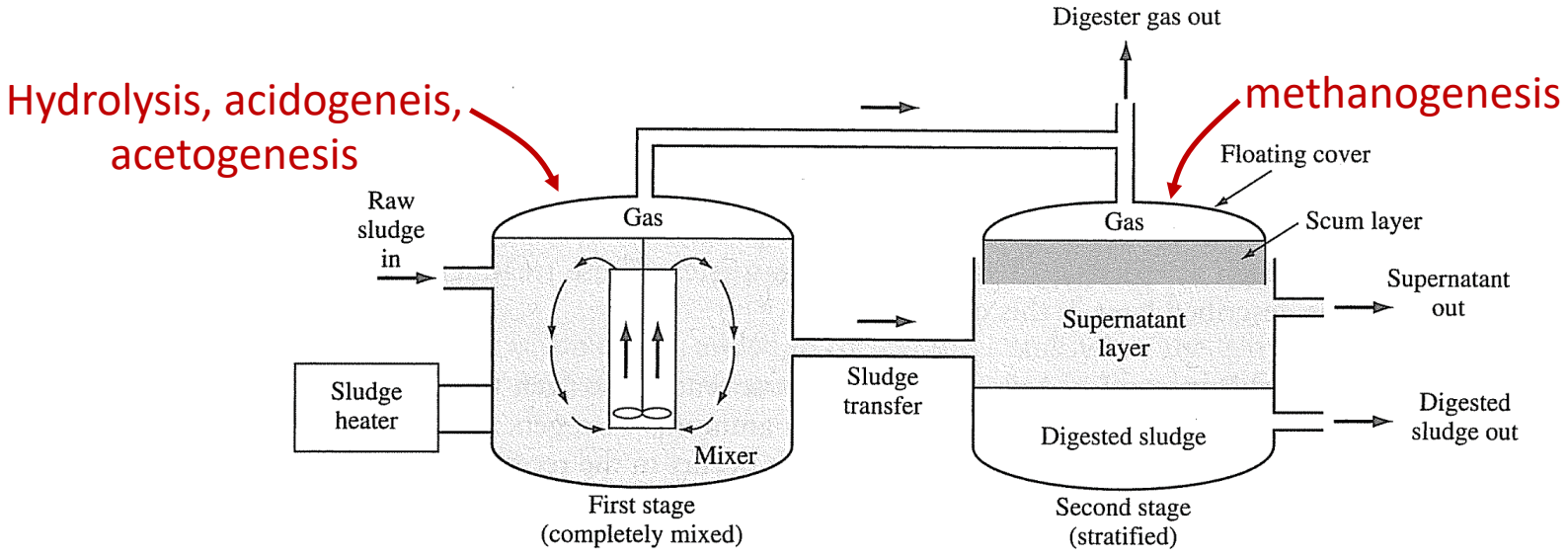
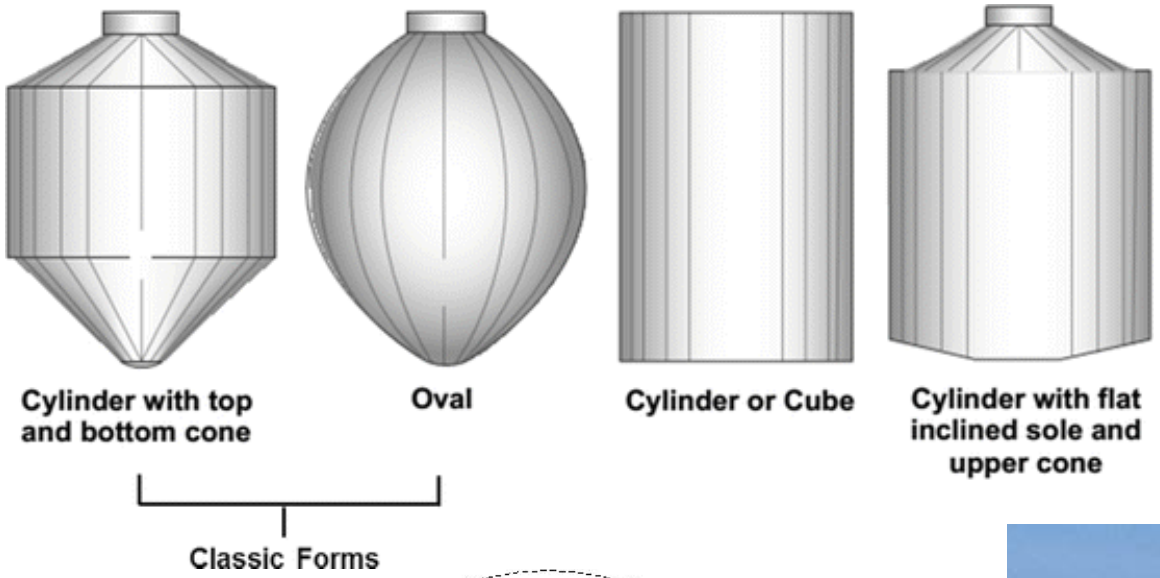
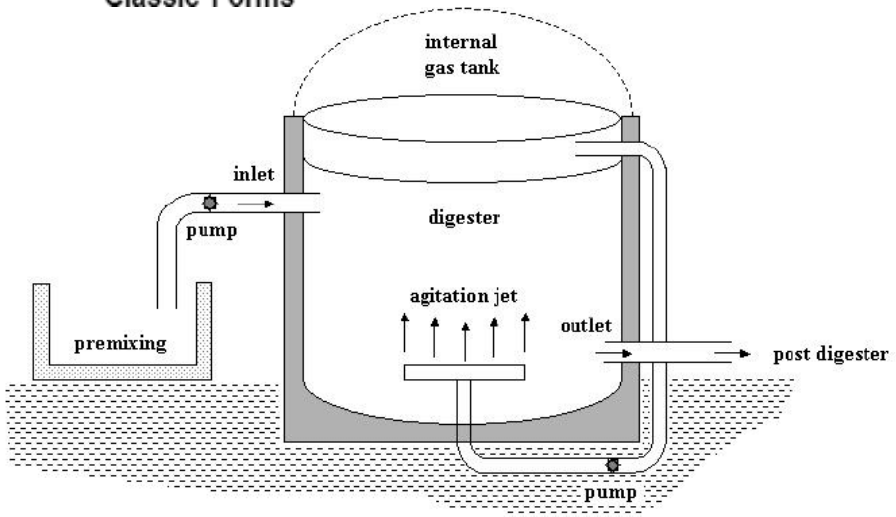


FIGURE 6.21 Schematic of a two-stage anaerobic digester.

Sludge treatment









(source: https://blog.anaerobic-digestion.com/anaerobic-digestion-in-the-us-northwest/; https://www.researchgate.net/figure/Digester-for-wet-anaerobic-digestion-European-example-slurry-organic-waste-crop_fig2_228792225; Radetic, 2018, Handbook of Water and Used Water Purification; https://www.water-technology.net/projects/reading_sewage/)

Readings and practice questions

- Readings
  - Essential reading: pp316-333

- Practice questions
  - Q6.28, 6.30